The Thiele Machine

A Complete Mathematical Specification

Derived from the Fully Machine-Checked Coq Corpus

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Abstract

This is the complete mathematical specification of the Thiele Machine: a formal model of computation where structural information carries an explicit, monotone cost μ that starts at zero and never decreases. Everything here is built from the machine-checked Coq corpus. You can reconstruct the machine from what's in this document alone.

Coq establishes that conclusions follow from axioms. It does *not* establish that axioms model physical reality. That distinction matters and I keep it explicit throughout. Every result is tagged as one of three types: **(S)** structural theorems about the machine as a mathematical object, **(C)** conditional derivations where a physics conclusion follows from a stated axiom, and **(R)** consistency relations that check internal coherence but are not independent predictions.

The core claims are structural (S): No Free Insight; μ-conservation; μ-hierarchy; μ-ledger necessity (five-way independence classification); categorical separation; strict Turing subsumption; the classical CHSH bound ($|S| \leq 2$); the algebraic Tsirelson bound ($|S| \leq 2\sqrt{2}$); and the zero-axiom biconditional `column_contractive_iff_quantum_realizable`. The CHSH algebraic result is pure math. The Landauer connection is motivation. Geometry: only the discrete angle-defect identity for planar triangulations.

The document also covers, with explicit epistemological tags, conditional derivations under named physical axioms (Born rule under linearity, no-cloning, unitarity) and reference explorations (4D simplicial-complex differential geometry, self-reference tower). These are marked (C) or (R) and do not form part of the primary claim ledger. There are exactly two named trust boundaries: `bsc_kami_compilation_trusted` (Layer 2 Verilog semantics gap) and `A_QM` (physical quantum axiom).

Contents

Chapter 1

Introduction

The Thiele Machine makes the cost of structure explicit. A Turing Machine treats all state transitions as cost-equivalent. The Thiele Machine assigns a specific cost μ to every operation that extracts or asserts structural information. That's the design. Here is where it goes.

The central claim, proven formally: *nothing gains structural insight about a system without paying μ-cost.* Combined with explicit physical axioms (linearity, information conservation), that constraint produces structural results that parallel quantum mechanics and thermodynamics. Each result's status, whether a structural theorem **(S)**, a conditional derivation **(C)**, or a consistency relation **(R)**, is marked directly. Nothing is hidden.

The document is organized as:

1. **Foundations**: State space, instruction set, operational semantics.
2. **Core Theorems**: No Free Insight, μ-ledger necessity, μ-Chaitin, μ-conservation.
3. **Computation**: Strict Turing subsumption, oracle pricing as instantiation of universal NFI, μ-complexity classes, μ-cost hierarchy.
4. **Symmetry & Conservation**: Gauge action, Noether's theorem, receipts.
5. **Quantum / CHSH algebra**: Tsirelson bound (algebraic), CHSH classical bound, biconditional column-contractive $\iff$ NPA-realizable, three-tier observation taxonomy. Pure polynomial arithmetic, zero physics axioms.
6. **Conditional physics**: Born rule (under linearity), unitarity, no-cloning, purification, Landauer (with second-law assumed as a record field), affine EFE family. Each is conditional on stated axioms in its own file.
7. **Discrete Geometry**: Discrete angle-defect identity for planar triangulations (not classical Gauss-Bonnet); 4D simplicial complex and Einstein-tensor definitions (reference exploration only, vacuum case is a $0 = 0$ consistency check).
8. **Meta-Theory**: Three-layer Coq/Python/RTL observable contract, categorical structure.
9. **Predictions**: Falsifiable predictions tied to specific Coq-verifiable claims.

1.1 Epistemological Framework

Formal verification (Coq) establishes that conclusions follow logically from axioms. It does **not** establish that axioms model physical reality. This distinction is critical for evaluating the physics claims in Parts V–VIII.

Every result in this specification falls into one of three categories:

(S) Structural Theorems about the Thiele Machine as a mathematical object. These are unconditionally true given the definitions. *Examples*: μ-conservation, gauge invariance, confluence, halting undecidability, strict Turing subsumption.

(C) Conditional Theorems of the form "if axiom X holds, then physics result Y follows." The logical inference is verified; whether axiom X holds in nature is an empirical question. *Examples*: Born rule uniqueness (conditional on linearity), complex necessity (conditional on 2D amplitudes), unitarity (conditional on information conservation).

(R) Consistency Algebraic identities or definitional equivalences that verify internal coherence but do not constitute independent predictions. *Example*: the Tsirelson cost definition.

Each major theorem in Parts V–VIII is annotated with its category. A theorem marked **(C)** or **(R)** is not diminished. Conditional derivations and consistency checks are valuable scientific tools. But the reader should not mistake them for derivations from first principles without stated assumptions. See Appendix B for the complete classification table.

Chapter 2

The State Space

2.1 Primitives

Let $\mathbb{N} = \{0, 1, 2, \ldots\}$. Let $\mathbb{B} = \{0, 1\}$. Let Σ be a finite alphabet.

2.2 Regions and Modules

Definition 2.1 (Region)**.** A **Region** R is a finite subset of $\mathbb{N}$, canonically represented as a deduplicated list preserving first-occurrence order: $\mathrm{norm}(L) = \mathrm{nodup}(L)$ removes duplicate elements while preserving the order of their first appearance. (The Python VM applies a stricter canonical form via `sorted(set(region))`; the Coq specification uses `nodup Nat.eq_dec` which does not sort.)

Definition 2.2 (Module)**.** A **Module** $M = (R_M, A_M)$ consists of a normalized region $R_M \subset \mathbb{N}$ and a set of axioms $A_M \subset \Sigma^*$.

2.3 The Partition Graph

Definition 2.3 (Partition Graph)**.** A partition graph $G = (N_{id}, \mathcal{T})$ where $N_{id} \in \mathbb{N}$ is the next available module ID and $\mathcal{T} : \{0, \ldots, N_{id} - 1\} \rightharpoonup \mathcal{M}$ is a finite partial function from IDs to modules.

Axiom 2.1 (Well-Formedness)**.** G is well-formed iff $\mathrm{dom}(\mathcal{T}) \subseteq \{0, \ldots, N_{id} - 1\}$.

2.4 Machine State

Definition 2.4 (VM State)**.** The complete state is a 12-tuple:

$$S = (G, C, R, \mathit{Mem}, PC, \mu, T_\mu, \mathit{err}, \ell, m, w, \mathit{cert})$$

where $G \in \mathcal{G}$ is the partition graph, $C \in \mathbb{N}^4$ comprises the CSRs (certificate address, status, error code, heap base), $R \in \mathbb{N}^{\texttt{REG\_COUNT}}$ is the register file, $\mathit{Mem} \in \mathbb{N}^{\texttt{MEM\_SIZE}}$ is main memory (the kernel constants `REG_COUNT` and `MEM_SIZE` are parametric; the synthesized RTL fixes `REG_COUNT` $= 16$ and `MEM_SIZE` $= 128$), $PC \in \mathbb{N}$ is the program counter, $\mu \in \mathbb{N}$ is the μ-ledger, $T_\mu \in \mathbb{N}^{16}$ is the flattened 4×4 μ-tensor (row-major), $\mathit{err} \in \mathbb{B}$ is the global error flag (latching), $\ell \in \mathbb{N}$ is the logic-engine accumulator, $m \in \mathbb{N}$ is the machine status register ($0 =$ Turing mode, $1 =$ Thiele mode), $w \in \mathbb{N}^8$ is the WitnessCount record (`vm_witness`): eight natural-number counters `wc_same_ab` and `wc_diff_ab` for each setting pair $(a, b) \in \{00, 01, 10, 11\}$, and $\mathit{cert} \in \mathbb{B}$ is the certified flag (`vm_certified`).

Chapter 3

Instruction Set and Cost Model

3.1 Instruction Categories

The instruction set $\mathcal{I}$ is organized into six families. Every instruction i carries an explicit cost parameter $\Delta\mu_i \in \mathbb{N}$.

3.1.1 Structural Operations

These modify the partition graph G:

- `PNEW`$(R, \Delta\mu)$: Create module with region R.
- `PSPLIT`$(m, L, R, \Delta\mu)$: Split module m into disjoint regions.
- `PMERGE`$(m_1, m_2, \Delta\mu)$: Merge two modules.
- `PDISCOVER`$(m, E, \Delta\mu)$: Discovery accounting for module m. The abstract specification attaches evidence E (list of axioms) to the module; the current hardware-aligned step advances PC and charges $\Delta\mu$ without mutating the graph.
- `MDLACC`$(m, \Delta\mu)$: Minimum-description-length accumulate on module m.

3.1.2 Logical Operations

These interact with information content:

- `LASSERT`$(\mathit{freg}, \mathit{creg}, \mathit{kind}, \mathit{flen}, \mathit{cost})$: Assert a formula read from `vm_mem`[`R[freg]`] (encoded-unit count *flen*, eight concrete bits per unit) against a witness package at `vm_mem`[`R[creg]`]. In SAT mode the package contains one satisfying assignment and one falsifying assignment, stored back-to-back; $\Delta\mu = \mathit{flen} \times 8 + S(\mathit{cost}) \geq 1$.
- `LJOIN`$(\mathit{c1reg}, \mathit{c2reg}, \Delta\mu)$: Join certificates from two memory-resident strings addressed by registers.
- `REVEAL`$(m, n, \mathit{cert}, \Delta\mu)$: Reveal n bits of structure from m; cost is $n + S(\Delta\mu)$.
- `EMIT`$(m, p, \Delta\mu)$: Emit payload p from module m; cost is $|p|_{\text{bits}} + S(\Delta\mu)$.

3.1.3 Computational Operations (Reversible ALU)

Reversible register/memory operations:

- `XFER`$(\mathit{dst}, \mathit{src}, \Delta\mu)$: Copy register *src* to *dst*.

- `XOR_LOAD`($dst, addr, \Delta\mu$): Load memory word at $addr$.
- `XOR_ADD`($dst, src, \Delta\mu$): $dst \leftarrow dst \oplus src$.
- `XOR_SWAP`($a, b, \Delta\mu$): Swap registers a and b.
- `XOR_RANK`($dst, src, \Delta\mu$): $dst \leftarrow \text{popcount}(src)$.

3.1.4 Standard ALU and Control Flow

- `LOAD_IMM`($dst, imm, \Delta\mu$): Load 32-bit immediate into register.
- `LOAD`($dst, addr, \Delta\mu$): Load from main memory.
- `STORE`($addr, src, \Delta\mu$): Store to main memory.
- `ADD`($dst, r_1, r_2, \Delta\mu$): $dst \leftarrow r_1 + r_2 \pmod{2^{64}}$.
- `SUB`($dst, r_1, r_2, \Delta\mu$): $dst \leftarrow r_1 - r_2 \pmod{2^{64}}$.
- `JUMP`($target, \Delta\mu$): Unconditional jump.
- `JNEZ`($r, target, \Delta\mu$): Jump if $R[r] \neq 0$.
- `CALL`($target, \Delta\mu$): Push return address, jump to target.
- `RET`($\Delta\mu$): Pop return address, jump back.

3.1.5 Auxiliary Operations

- `CHSH_TRIAL`($x, y, a, b, \Delta\mu$): Run a CHSH correlation trial; $x, y, a, b \in \{0, 1\}$ required.
- `CHECKPOINT`($label, \Delta\mu$): Execution checkpoint (charges μ).
- `READ_PORT`($dst, ch, val, bits, \Delta\mu$): Read from I/O channel (cert-setter).
- `WRITE_PORT`($ch, src, \Delta\mu$): Write to I/O channel.
- `HEAP_LOAD`($dst, addr, \Delta\mu$): Load from heap-relative address.
- `HEAP_STORE`($addr, src, \Delta\mu$): Store to heap-relative address.
- `HALT`($\Delta\mu$): Halt execution.

3.1.6 Categorical Morphism Operations

These 7 opcodes (0x27–0x2D) implement the categorical layer over partition modules. See Chapter 23 for laws and proofs.

- `MORPH`($dst, src, tgt, idx, \Delta\mu$): Create morphism $src \to tgt$; write morphism ID to `regs`[dst].
- `COMPOSE`($dst, f, g, \Delta\mu$): Compose $f; g$ when f.tgt $= g$.src; errors on type mismatch.
- `MORPH_ID`($dst, m, \Delta\mu$): Create identity morphism id_m (diagonal relation).
- `MORPH_DELETE`($f, \Delta\mu$): Remove morphism f; cascades when modules are split/merged.
- `MORPH_ASSERT`($f, prop, cert, \Delta\mu$): **Cert-setter.** Charges $S(\Delta\mu) \geq 1$ unconditionally. Even failed attempts (morphism does not exist) incur cost 1. This is the strongest form of the NoFI law: the machine refuses to let even a doomed attempt at certifying a relational claim slip through for free.

- MORPH_TENSOR$(dst, f, g, \Delta\mu)$: Tensor product $f \otimes g$ (requires union modules to exist).
- MORPH_GET$(dst, f, sel, \Delta\mu)$: Read morphism metadata: selector $0 =$ src, $1 =$ tgt, $2 =$ coupling length, $3 =$ is_identity.

The active implementation exposes a 51-opcode ISA: 14 arithmetic/data, 8 memory and control flow, 7 I/O and tensor, 4 partition operations, 3 logic and certification, 8 witness and certification (CHSH_TRIAL, CERTIFY, REVEAL, CHSH_LASSERT, and the four Q_{1+AB} check variants), 7 categorical morphism operations $(14+8+7+4+3+8+7 = 51)$. 47 of the 51 are synth-realized; the four Q_{1+AB} variants live in the Kami HW abstraction with kernel-equivalence proven but are excluded from the synthesized Verilog by silicon budget. This condensed mathematical specification presents the core instruction families relevant to the kernel semantics and cost model. Every instruction carries an explicit $\Delta\mu$ parameter; the function instruction_cost extracts it.

3.2 Cost Model

Every instruction i carries an explicit cost parameter $\Delta\mu_i \in \mathbb{N}$. The function instruction_cost(i) extracts this parameter. The cost is applied via apply_cost$(s, i) = s.\mu +$ instruction_cost(i). Since $\Delta\mu_i : \mathbb{N}$, the μ-ledger is monotonically non-decreasing by construction.

The mandatory positive-cost class is: CERTIFY, MORPH_ASSERT, and LJOIN (each charges $S(\Delta\mu) \geq 1$); EMIT $(|payload|_{\text{bits}} + S(\Delta\mu))$; REVEAL and READ_PORT $(bits + S(\Delta\mu))$; and LASSERT $(flen \times 8 + S(\Delta\mu))$. These cannot be zero-cost even with $\Delta\mu = 0$. All other instructions charge exactly their encoded $\Delta\mu$, which is 0 in typical programs.

Chapter 4

Operational Semantics

4.1 Transition Function

The transition $\delta : \mathcal{S} \times \mathcal{I} \to \mathcal{S}$ is defined by:

1. $PC' = PC + 1$ for non-control-flow instructions; JUMP, JNEZ, CALL, and RET set PC to the target address; HALT stops execution.
2. $\mu' = \mu + \Delta\mu_i$.
3. $err' = err \vee \mathrm{fail}(S, i)$.

4.2 Traces and Execution

Definition 4.1 (Trace)**.** A trace $\tau = [i_0, \ldots, i_k] \in \mathcal{I}^*$ is a finite sequence of instructions.

Definition 4.2 (Execution)**.** $\mathrm{Run}([], S) = S; \quad \mathrm{Run}(i :: \tau', S) = \mathrm{Run}(\tau', \delta(S, i))$.

4.3 Selected Instruction Semantics

PNEW: Normalize region; if already present, idempotent; else add $(R_{\mathrm{norm}}, \emptyset)$ to G.

REVEAL: Update CSRs with checksum of certificate *cert*. Cost: $bits + S(\Delta\mu)$.

LASSERT: Read formula from `vm_mem`[`R[formula_addr_reg]`] (encoded-unit count `formula_len`, eight concrete bits per unit) and witness package from `vm_mem`[`R[cert_addr_reg]`]. `cert_kind = true` (SAT): verify using a satisfying model plus a falsifying countermodel (dual-witness). `cert_kind = false` (UNSAT): the kernel step always fails at this path—`lassert_check_ok` returns `false` unconditionally; LRAT checking is not invoked in the step semantics (documented gap in `VMStep.v`). If valid, the constraint is certified; if invalid, set $err \leftarrow$ true. Note: in the current hardware-aligned semantics, LASSERT validates the constraint but does not directly mutate the partition graph's axiom set — cert-channel activation happens through the subsequent CERTIFY or MORPH_ASSERT. Cost: `formula_len` $\times 8 + S($`mu_delta`$)$.

Chapter 5

No Free Insight

The central theorem: structural insight is never free.

5.1 Definitions

Definition 5.1 (Receipt Predicate)**.** $P : \mathcal{O} \to \mathbb{B}$ is a computable function on observation histories.

Definition 5.2 (Strength Ordering)**.** $P_1 \leq P_2$ iff $\forall o.\ P_1(o) = 1 \implies P_2(o) = 1$. $P_1 < P_2$ iff $P_1 \leq P_2$ and $\exists o.\ P_2(o) = 1 \wedge P_1(o) = 0$.

Definition 5.3 (Certification)**.** Trace τ **certifies** P iff execution succeeds ($err = 0$), the supra-certificate flag is set, and the output satisfies P.

Definition 5.4 (Structure Addition)**.** HasStructureAddition(τ) is true iff τ contains a `REVEAL`, `LASSERT`, or equivalent operation.

5.2 The Theorem

Theorem 5.1 (No Free Insight (NoFreeInsight_Theorem.v))**.** *Let $P_{\text{strong}} < P_{\text{weak}}$. If a trace τ starting from a clean state s_0 certifies P_{strong}, then:*

$$\text{Certified}(\tau, P_{\text{strong}}) \implies \text{HasStructureAddition}(\tau)$$

Proof Sketch. The proof is parametric: `NoFreeInsight_Theorem.v` proves the theorem for any system satisfying the `NO_FREE_INSIGHT_SYSTEM` module type interface. The interface requires four contract obligations (not Coq `Axiom` declarations—they are proved separately for the kernel instantiation): (1) μ-monotonicity, (2) a certification specification linking `certifies` to execution, (3) a no-free-insight contract stating that strict strengthening requires structure addition, and (4) that the system has a notion of clean start. The theorem file itself contains zero `Axiom`, zero `Admitted`. The contract obligations are discharged by the kernel instantiation. □

Corollary 5.2 (Cost of Insight)**.** *Gaining insight (strengthening the predicate) implies $\Delta\mu > 0$.*

5.3 μ-Ledger Necessity

Theorem 5.3 (μ-Ledger Necessity and Complete Independence Classification (NecessityOfMuLedger.v, kernel/NecessityAbstract.v) — **(S)**)**.** *Five independence results are formally proven, zero Admitted:*

1. *$\mu \perp (\mathtt{mem}, \mathtt{regs}, \mathtt{pc})$: no strict-classical oracle recovers μ* `turing_ram_mu_necessity`

2. $\texttt{certified} \perp (\texttt{mem}, \texttt{regs}, \texttt{pc})$: *no strict-classical oracle recovers certification* `turing_ram_cert_necessity`

3. $\texttt{certified} \perp (\texttt{mem}, \texttt{regs}, \texttt{pc}, \mu)$: *knowing μ does not determine certification* `cost_model_cert_necessity`

4. $\mu \perp (\texttt{mem}, \texttt{regs}, \texttt{pc}, \texttt{certified})$: *knowing certification does not determine μ* `cert_model_mu_necessity`

5. $\texttt{vm\_graph} \perp (\texttt{mem}, \texttt{regs}, \texttt{pc}, \mu, \texttt{certified})$: *the categorical layer is independent of the full μ-ledger shadow* `graph_not_recoverable_from_P_full`

Three-component independence *(`thiele_state_three_component_independence`): μ, `vm_certified`, and `vm_graph` are the three provably independent non-classical state components. Each is irrecoverable from any combination of the others plus $(\texttt{mem}, \texttt{regs}, \texttt{pc})$.*

Minimality *(`P_full_is_minimal_complete_extension`): $(\texttt{mem}, \texttt{regs}, \texttt{pc}, \mu, \texttt{certified})$ is the minimal complete extension of classical state for μ and certification. Graph structure is a third independent axis.*

Universality *(`mu_ledger_necessity_universal`): The separation holds from* every *VMState, not only from crafted zero-state witnesses. From any reachable state, `CERTIFY 0` and `PNEW [] 0` produce identical strict classical shadows but different μ and certification.*

Trace closure *(`po1_trace_necessity`): For any common prefix program of any length, the two witness extensions maintain equal strict shadows but strictly different μ. No accumulation of classical computation closes the gap.*

Proof Sketch. Each separation uses explicit witnesses where equal projected inputs force equal oracle outputs, while the witness states require unequal outputs, a contradiction. Strict shadow equality: both `CERTIFY 0` and `PNEW [] 0` advance `pc` by 1 and preserve `mem`/`regs`. μ divergence: CERTIFY charges $S(0) = 1$; PNEW charges 0. Graph separation uses `categorical_separation` from `PartitionSeparation.v`: two states with identical P_{full} but different `pg_morphisms`. Universality follows because `vm_apply_certify_strict_shadow` and `vm_apply_pnew_strict_shadow` hold for arbitrary source states. ☐

Theorem 5.4 (Latent μ: Cost is Intrinsic to Computation (NecessityOfMuLedger.v §7) — **(S)**). *The μ-cost accumulated by any computation is a property of the instruction sequence, not of any tracking system. Four results, all zero Admitted:*

1. `shadow_mu_is_computation_intrinsic`: *For any program* `prog` *and any starting state s:*

$$(\texttt{run\_instrs}\ s\ \texttt{prog}).\mu = s.\mu + \texttt{trace\_total\_cost}(\texttt{prog})$$

   The increment is determined entirely by the instruction sequence, independent of vm_graph, vm_regs, vm_mem, vm_certified, and the starting μ.

2. `shadow_mu_delta_universal`: *Two machines running the same program from any two starting states accumulate the same additional μ. The cost is written in the program before execution begins.*

3. `shadow_mu_unique_accounting`: *Any instruction-consistent accounting system M satisfying $M(s) = 0$ at the starting state assigns exactly* `trace_total_cost` *to the result. Replace the Thiele Machine's μ-counter with any other honest accounting and you get the same number.*

4. `shadow_mu_inevitable`: *Any program containing an instruction with positive declared cost accumulates strictly positive μ from any starting state. There is no free computation.*

Interpretation. *A Turing machine executing a program pays exactly* `trace_total_cost` *in latent μ on every execution. The cost was always there, forced by the instruction sequence, but the Turing machine had no mechanism to record or expose it. The μ-ledger does not create this cost. It reveals what every computation was already paying.*

5.4 Partition Refinement and No Free Insight

Theorem 5.5 (Partition Ops Are Free (kernel/PartitionRefinementNoFI.v) — **(S)**). *PNEW, PSPLIT, and PMERGE are not cert-setters and can carry* $\Delta\mu = 0$*:*

- `partition_structural_ops_not_cert_setters`*: None of the three partition operations touch* `csr_cert_addr` *or* `vm_certified`*.*
- `partition_structural_ops_can_be_free`*: All three can execute with* $\Delta\mu = 0$*.*
- `partition_structural_trace_cannot_certify`*: A trace containing only PNEW, PSPLIT, and PMERGE (all at* $\Delta\mu = 0$*) cannot reach a certified state from an uncertified start.*

Building partition structure costs nothing by itself. Certifying structural claims about that partition is what costs.

Theorem 5.6 (Partition Free But Certification Non-Free (kernel/PartitionRefinementNoFI.v) — **(S)**). `partition_free_but_certification_nonfree`*: There exist traces where all partition-structural instructions carry* $\Delta\mu = 0$ *(so* $\mu_{\text{final}} = \mu_{\text{init}} = 0$*) yet the trace ends with a partition graph that a subsequent MORPH_ASSERT or CERTIFY can certify — but only at* $\Delta\mu \geq 1$*. Partition structure can be built at zero* μ*-cost; certified partition knowledge cannot be acquired for free. Zero Admitted.*

Theorem 5.7 (Partition Refinement Non-Free (kernel/PartitionRefinementNoFI.v) — **(S)**). `partition_refinement_nonfree`*: Any trace that starts with no cert evidence and ends with a certified claim about a partition structure paid at least 1 in* μ*. This is the NoFI law specialized to partition refinement. Zero Admitted.*

Chapter 6

The μ-Chaitin Theorem

A Chaitin-style incompleteness theorem denominated in μ-currency.

Theorem 6.1 (μ-Chaitin (MuChaitinTheory_Theorem.v))**.** *For any formal theory system T satisfying the μ-Chaitin interface, the number of bits k the theory can certify is bounded by:*

$$k \leq |T| + c$$

where $|T|$ is the description length of the theory and c is a fixed overhead constant. The proof chain: $k \leq \text{payload} \leq \mu\text{-}\mathit{info} \leq \text{budget} \leq |T| + c$.

Chapter 7

Computational Universality and Subsumption

7.1 Strict Containment

Theorem 7.1 (Turing $\subsetneq$ Thiele (kernel/Subsumption.v)). *Classical Turing computation is* ***properly extended*** *by sighted Thiele computation (both are Turing-complete, but Thiele carries additional partition and μ-accounting structure):*

$$\exists\, p.\ \mathrm{sighted}(p) \wedge \neg\mathrm{turing}(p)$$

There exist programs that are Thiele-computable (using partition structure and μ-accounting) but are not classical Turing programs. The extra structure is genuine new content.

7.2 Oracle Cost Lower Bound

Theorem 7.2 (Oracle pricing self-consistency (conceptual; not separately formalized)). *Inside the No Free Insight framework, the universal lower bound on certification cost (`universal_nfi_any_substrate` in `coq/kernel/nfi/UniversalCertificationCost.v`) immediately implies that any sound oracle satisfying the one-step rule `cs_cert_costs` (cost ≥ 1 on any uncertified-to-certified transition) charges $\geq n$ for n certifying queries. There is no separate oracle-impossibility file in the codebase; the oracle case is a direct instantiation of the universal theorem at a `CertificationSystem` whose state encodes oracle responses. The physical justification for charging $\mu > 0$ on oracle queries comes from NoFI: gaining certified information about undecidable propositions has positive μ-cost.*

7.3 μ-budget predicates

Definition 7.1 (μ-budget predicates (kernel/MuComplexity.v)). Let $p_{\mathrm{fuel}}, p_\mu : \mathbb{N} \to \mathbb{N}$ be polynomial bounds and $\mathrm{size} : \mathrm{VMState} \to \mathbb{N}$ a size function.

- A decision problem f is `mu_budget_decidable` (with respect to p_{fuel} and p_μ) iff a solver trace exists for every input: a trace of length $\leq p_{\mathrm{fuel}}(\mathrm{size}(s))$ using at most $p_\mu(\mathrm{size}(s))$ units of μ-cost that runs without error.

- A decision problem f is `mu_budget_verifiable` iff every positive instance has a certificate trace verifiable within the same fuel and μ-cost bounds, ending with `vm_certified = true`.

Theorem 7.3 (Zero-μ classical solvers are μ-budget decidable (kernel/MuComplexity.v) — **(S)**)**.** *`classical_mu_budget_decidable`: any decision problem solved by a zero-μ classical program within polynomial fuel is `mu_budget_decidable` with μ-bound 0. Classical computation is a degenerate case of μ-computation. Zero Admitted.*

Remark 7.1. `mu_budget_decidable` and `mu_budget_verifiable` are predicates classifying decision problems by their polynomial-μ envelope, not classical complexity classes. The Coq file proves the basic inclusion (every decidable problem is verifiable, by re-running the solver and certifying). It does not prove a class separation. The factored-search witness in `StructuralAdvantage.v` gives a concrete program-level μ-cost advantage (paying $\mu = 18$ to certify a decomposition and reduce search from $O(N^2)$ to $O(2N)$), but that is not a class separation. Earlier versions of this section named these predicates `in_muP` and `in_muNP`; the names were misleading because they evoked the classical P-versus-NP question, which this file does not address.

Theorem 7.4 (μ-Cost Hierarchy (kernel/MuHierarchyTheorem.v) — **(S)**)**.** *For every $k \geq 1$, the μ-dimension admits a strict hierarchy:*

1. ***Achievability**: there exists a trace from `init_state` with μ-cost exactly k that is* level-k certified *— the run executes a `CERTIFY` δ instruction with $S(\delta) \geq k$ and ends with `vm_certified = true`.*
2. ***Lower bound**: any level-k certified trace from `init_state` must have spent at least k units of μ-cost.*

Formally (`mu_hierarchy_theorem`): for every $k \geq 1$, the witness trace [`instr_certify`$(k-1)$] (cost $= S(k-1) = k$) executes a level-k `CERTIFY` from `init_state`; conversely, the trace-side predicate `level_k_certified` forces `trace_mu_cost` $\geq k$ via `level_k_certification_cost_floor`, since the `CERTIFY`'s declared cost appears in the executed-instruction ledger and the ledger sum equals the trace cost (`run_vm_mu_conservation`).

***Corollary** (`mu_hierarchy_no_upper_bound`): no fixed μ-budget suffices for all levels. The μ-dimension is unbounded and non-collapsible. Zero Admitted.*

Remark 7.2 (μ-Hierarchy as a P-vs-NP analogue, not a P-vs-NP result)**.** Theorem 7.4 is the μ-analogue of the classical time hierarchy theorem. It is not a statement about classical P versus NP, which this project does not touch. In the certified-structure setting it makes one thing precise: *finding* a certificate of structural complexity k costs $\mu \geq k$, while *verifying* it (given the trace) costs 0 extra μ. The hierarchy is strict and unbounded, with zero `Admitted`, no oracle, and no diagonalization. The lower-bound proof is trace-side: a level-k certified run executed a specific `CERTIFY` δ with $S(\delta) \geq k$, and `executed_instruction_cost_recorded` plus `ledger_sum_contains_lower_bound` place that cost in the ledger sum, which `run_vm_mu_conservation` equates with the trace's μ-cost.

7.4 Intrinsic level hierarchy

Definition 7.2 (Intrinsic level (coq/IntrinsicLevelHierarchy.v))**.** A final state s is *intrinsically certifiable at level* $\geq k$, written `level_intrinsic_at_least`(k, s), iff every `run_vm` trace from `init_state` that ends at s with `vm_certified = true` executes at least k cert-setter instructions:

$$\forall \mathit{fuel}\ \mathit{trace}.\ \texttt{run\_vm}(\mathit{fuel}, \mathit{trace}, \texttt{init\_state}) = s \ \Rightarrow\ s.\texttt{vm\_certified} = \texttt{true} \ \Rightarrow\ \texttt{cert\_setter\_executions}(\mathit{fu}$$

The level is a property of the *final state*, not of any particular trace reaching it.

Theorem 7.5 (Intrinsic level forces μ (coq/IntrinsicLevelHierarchy.v::level_intrinsic_forces_mu))**.** *If* `level_intrinsic_at_least`(k, s) *holds and* $s.$`vm_certified` $=$ `true`*, then for every trace reaching* s *from* `init_state`*,* $s.$`vm_mu` $\geq k$*. Zero Admitted.*

Theorem 7.6 (Strict separation (coq/IntrinsicLevelHierarchy.v::level_strict_separation))**.** *For every* $k \geq 0$*, if* `level_intrinsic_at_least`$(k+1, s)$ *holds and* $s.$`vm_certified` $=$ `true`*, then every trace reaching* s *executes strictly more than* k *cert-setter instructions. Zero Admitted.*

Remark 7.3 (Relation to Theorem 7.4)**.** Theorem 7.6 is the state-level counterpart of the trace-level μ-hierarchy. The trace-level version (`mu_hierarchy_theorem`, `level_k_certification_cost_floor`) defines level on the trace — a trace is level-k certified iff it executes a `CERTIFY` δ with $S(\delta) \geq k$. The state-level version defines level on the final state — every trace reaching it requires at least k cert-setter executions. The two are complementary entry points to the same hierarchy: the trace-level theorem gives the achievability witness ([`instr_certify`$(k-1)$] from `init_state`); the state-level theorem gives the lower bound on μ for any state intrinsically requiring k cert-events. Both are present-tense in the kernel; both close under the global context.

7.5 Verifier model and the bare-setting impossibility

Definition 7.3 (Bare verifier (coq/VerifierModel.v))**.** The bare-setting transcript type is `BareTranscript := list StrictClassicalState`. A bare verification problem is a record `BareVerificationProblem` carrying

- `bare_honest_claim` : `VMState` $\rightarrow$ `Prop`;
- `bare_explains` : `VMState` $\rightarrow$ `BareTranscript` $\rightarrow$ `Prop` (the transcript-honestly-witnessed-by-state relation);
- `bare_cheap_budget`, `bare_recompute_cost` : $\mathbb{N}$.

A bare verifier is a record `BareVerifier` with `bv_decide` : `BareTranscript` $\rightarrow$ `bool` and `bv_cost` : `BareTranscript` $\rightarrow \mathbb{N}$.

Definition 7.4 (Soundness, completeness, cheapness (coq/VerifierModel.v))**.** For a problem P and verifier V:

- `bare_sound`$(P, V) := \forall t.\ V(t) =$ `true` $\Rightarrow \forall s.$ `bare_explains`$(P, s, t) \Rightarrow$ `bare_honest_claim`(P, s).
- `bare_complete`$(P, V) := \forall s\, t.$ `bare_honest_claim`$(P, s) \Rightarrow$ `bare_explains`$(P, s, t) \Rightarrow V(t) =$ `true`.
- `bare_cheap`$(P, V) := \forall t.$ `bv_cost`$(V, t) \leq$ `bare_cheap_budget`(P).

The soundness condition is *strong*: the claim must hold for every state explaining the transcript, not merely for some.

Theorem 7.7 (Bare-setting impossibility (coq/VerifierImpossibility.v::bare_setting_no_sound_complete_verifi *There is no bare verifier simultaneously* `bare_sound` *and* `bare_complete` *on the verification problem* `mu_eq_one_problem` *whose claim is* `vm_mu`$(s) = 1$ *and whose* `bare_explains` *relation holds for the two* `NecessityOfMuLedger` *witness states against the shared strict-shadow trace.*

Proof. `po1_state_A` (the cost-1 `CERTIFY 0` result, $\mu = 1$) and `po1_state_B` (the cost-0 `PNEW [] 0` result, $\mu = 0$) project to identical strict-shadow traces. Completeness applied to A's honest run forces V to accept the shared transcript. Strong soundness applied to the same accepted transcript at the explanation by B forces `vm_mu`$(B) = 1$, contradicting `vm_mu`$(B) = 0$. `Print Assumptions` returns `Closed under the global context`. $\square$

Theorem 7.8 (Three sufficient escapes (coq/VerifierEscape_{Substrate,Hardness,Interaction}.v))**.** *Each kernel theorem below exhibits a constant-cost verifier that simultaneously satisfies soundness and completeness on the μ-sensitive claim by augmenting the transcript with non-classical structure:*

- `substrate_escape_succeeds` *— transcript is* `list VMState`*; verifier reads μ directly.*
- `hardness_escape_succeeds` *— transcript is* `BareTranscript` $\times$ `bool` *(commitment bit); verifier accepts the commitment under a parameterised* `HardnessHypothesis` *record.*
- `interactive_escape_succeeds` *— transcript is* `BareTranscript` $\times \mathbb{N}$ *(response); verifier checks the response against the claim.*

Each closes under the global context with zero `Admitted`*.*

Theorem 7.9 (Factorisation impossibility (coq/VerifierExhaustiveness.v::V_does_not_factor_through_classica *Let T be any transcript type with a projection $\pi : T \to$* `BareTranscript`*. For any verifier $V : T \to$* `bool` *simultaneously sound and complete on the μ-sensitive claim (with the lifted-explains relation), V does not factor through π: there exist $t_1, t_2 : T$ with $\pi(t_1) = \pi(t_2)$ and $V(t_1) \neq V(t_2)$.*

Remark 7.4 (The verifier corollary)**.** Sound and complete verification of μ-sensitive claims requires the transcript to carry non-classical structural information — exactly the structure the projection `strict_shadow` drops. The three escapes above are concrete realisations; Theorem 7.9 says no verifier on the bare classical projection alone succeeds, regardless of how rich its transcript is otherwise. The substrate channel is the option the structural axis makes available; hardness and interaction are the two routes classical cryptography and complexity already used.

Chapter 8

μ-Conservation and Receipt Integrity

8.1 The μ-Ledger Identity

Theorem 8.1 (μ-Ledger Conservation (MuLedgerConservation.v)). *For any trace $\tau = [i_0, \ldots, i_k]$:*

$$\mu_{\text{final}} = \mu_{\text{init}} + \sum_{j=0}^{k} \text{cost}(i_j)$$

Every instruction increases μ by exactly its declared cost. No axioms, no admits.

Corollary 8.2 (Monotonicity). *μ never decreases: $\mu(s) \leq \mu(\text{Run}(\tau, s))$ for all τ.*

Corollary 8.3 (Irreversibility Bound). $\text{irreversible\_bits}(\tau) \leq \mu_{\text{final}} - \mu_{\text{init}}$.

Theorem 8.4 (μ Decomposition (MuLedgerConservation.v)). *$\mu_{\text{total}} = \mu_{\text{blind}} + \mu_{\text{sighted}}$ for any partition of the cost into blind (reversible) and sighted (structural) components.*

8.2 Receipt Integrity

Definition 8.1 (Receipt). A receipt $r = (\text{step}, \text{instr}, \mu_{\text{pre}}, \mu_{\text{post}}, h_{\text{pre}}, h_{\text{post}})$ records one transition.

Theorem 8.5 (Receipt Integrity (ReceiptIntegrity.v)). *A valid receipt or execution attestation proves $\mu_{\text{final}} = \mu_{\text{init}} + \sum \text{costs}$ for the attested run. Any receipt claiming $\Delta\mu \neq \text{cost}(\text{instr})$ fails validation.*

Theorem 8.6 (Non-Forgeability (ReceiptIntegrity.v)). *Forged receipts (claiming incorrect $\Delta\mu$) fail validation. Overflow values ($\mu > 2^{31} - 1$) are rejected at the receipt layer.*

Chapter 9

Gauge Symmetry and Noether's Theorem

9.1 The $\mathbb{Z}$-Action (Gauge Group)

Definition 9.1 (Gauge Shift)**.** For $\delta \in \mathbb{Z}$, the gauge shift $\sigma_\delta : \mathcal{S} \to \mathcal{S}$ maps $S \mapsto S[\mu \leftarrow \mu + \delta]$, preserving all other fields.

Theorem 9.1 (Group Action (KernelNoether.v))**.** *The gauge shifts form a $\mathbb{Z}$-action on states:*

1. ***Identity**: $\sigma_0(S) = S$.*
2. ***Composition**: $\sigma_a(\sigma_b(S)) = \sigma_{a+b}(S)$.*
3. ***Inverse**: $\sigma_{-n}(\sigma_n(S)) = S$ (when $\mu + n \geq 0$).*

9.2 Gauge Invariance

Definition 9.2 (Observable)**.** $\mathrm{Obs}(S)$ extracts the partition region structure, ignoring μ.

Theorem 9.2 (Gauge Invariance (KernelNoether.v))**.** $\mathrm{Obs}(\sigma_\delta(S)) = \mathrm{Obs}(S)$*. Shifting μ by a constant does not affect observables.*

Theorem 9.3 (Noether's Theorem (`coq/kernel/curvature/KernelNoether.v`))**.**

1. ***Forward**: States identical except for μ lie on the same gauge orbit. (This is direct from the definition: if all other fields match, the μ difference is the gauge shift.)*
2. ***Gauge Invariance**: `z_gauge_invariance` proves that shifting μ does not change `Observable_partition`. This is definitional: `z_gauge_shift` only modifies the `vm_mu` field, and `Observable_partition` reads only `vm_graph`.*
3. ***Trace Preservation**: Gauge-equivalent states produce identical observable traces (same labels, same μ-costs) for any finite horizon. Proven as `vm_step_orbit_equiv`.*

The μ-ledger is a gauge degree of freedom. Its absolute *value is unobservable; only* differences *$\Delta\mu$ are physical. The "Noether" label is a deliberate analogy: the result follows from the record structure (separate fields for μ and graph), not from a deep symmetry principle.*

Theorem 9.4 (μ-Monotonicity (KernelNoether.v))**.** $\mathrm{vm\_step}(S, i, S') \implies \mu(S) \leq \mu(S')$*. The μ-ledger never decreases under the dynamics. This is true by construction: instruction costs are natural numbers (≥ 0), so adding them to the ledger cannot decrease it. The proof examines each step constructor and confirms the arithmetic.*

Remark 9.1 (Part V Scope: Conditional Results and Named Axioms)**.** The chapters in this part present results conditional on named physical axioms (linearity in the Bloch parameter for the Born rule, the second law for Landauer, and the `A_QM` bridge for quantum realizability). They are marked **(C)** or **(R)** throughout. The primary unconditional quantum results — the classical CHSH bound, the algebraic Tsirelson bound, and the biconditional `column_contractive_iff_quantum_realizable` — appear earlier (Chapters 12–13). Those are pure algebra, no physics axioms.

The results in this part are not primary claims of the project. They are explorations of what follows *if* you add specific physical axioms. They show how far the machine reaches without claiming any physics the proofs do not support.

Chapter 10

The Born Rule

Definition 10.1 (Bloch Sphere Probabilities)**.** For a state with purity vector (x, y, z):

$$P(|0\rangle) = \frac{1+z}{2}, \qquad P(|1\rangle) = \frac{1-z}{2}$$

Definition 10.2 (Measurement μ-Cost)**.**

$$\text{Cost}(x, y, z) = \frac{1 - (x^2 + y^2 + z^2)}{2}$$

This is the linear entropy. For pure states ($r^2 = 1$), cost $= 0$. For mixed states ($r^2 < 1$), cost > 0.

Theorem 10.1 (Uniqueness of the Born Rule (BornRule.v))**.** *The Born rule $P(|0\rangle) = (1+z)/2$ is the* ***unique*** *probability assignment satisfying:*

1. *Non-negativity and normalization: $P \geq 0$, $P(|0\rangle) + P(|1\rangle) = 1$.*
2. *Eigenstate consistency: $P(0, 0, 1) = 1$, $P(0, 0, -1) = 0$.*
3. *Linearity in z: $P(x, y, z) = az + b$ for some a, b.*

Proof: Boundary conditions force $a = 1/2$, $b = 1/2$. No other solution exists.

Note: *The Coq theorem also lists a* `receipt_mu_consistent` *hypothesis, but it is* ***unused*** *in the proof. The hypothesis* `receipt_mu_consistent` *P unfolds to "measurement cost ≥ 0 for valid Bloch vectors," which is directly from the definition true for all states and does not constrain P. The linearity assumption (item 3) does all the work—it assumes the functional form $P = az + b$ and solves a 2-equation system. This is less "deriving the Born rule from μ-accounting" and more "the Born rule is the unique linear rule consistent with eigenstates."*

Chapter 11

Unitarity

Definition 11.1 (Evolution)**.** An evolution E maps Bloch vectors $(x, y, z) \mapsto (x', y', z')$ with associated μ-cost.

Theorem 11.1 (Unitarity from Zero Cost (Unitarity.v))**.**

1. *$\mu = 0 \implies r'^2 \geq r^2$ (purity cannot decrease at zero cost). Proven as* `zero_cost_preserves_purity`*.*

2. *Non-unitary evolution ($r'^2 < r^2$ for some state) requires $\mu > 0$. Proven as* `nonunitary_requires_mu`*: the proof specializes the conservation hypothesis at the witness, then* `lra`*.*

3. *$\mu = 0 \Rightarrow$ no info loss (one direction proven). The converse and the full equivalence $\mu = 0 \Leftrightarrow$ unitary $\Leftrightarrow$ reversible are* stated *in the file but not proven as theorems: they appear as* `Definition` *declarations (propositional constants) rather than proven* `Theorem`*s.*

Theorem 11.2 (Lindblad Requires μ (Unitarity.v))**.** *Lindblad-type dissipation at rate $\gamma > 0$ requires $\mu \geq \gamma$. Decoherence is paid for.*

Theorem 11.3 (CPTP Structure (Unitarity.v))**.** *Positivity + trace preservation $\implies$ the evolution is CPTP (completely positive, trace-preserving).*

Chapter 12

Purification

Theorem 12.1 (Purification Principle (Purification.v))**.** *Every mixed state* (x, y, z) *with* $r^2 = x^2 + y^2 + z^2 \leq 1$ *admits eigenvalues* λ_1, λ_2 *with:*

- $\lambda_1 + \lambda_2 = 1, \quad 0 \leq \lambda_i \leq 1.$
- $(\lambda_1 - \lambda_2)^2 = r^2.$
- *Construction:* $\lambda_1 = (1 + r)/2$*,* $\lambda_2 = (1 - r)/2$*.*

Pure states ($r = 1$) *need no external reference (deficit* $= 0$*). The maximally mixed state* ($r = 0$) *has deficit* $= 1$*.*

Chapter 13

No-Cloning

Theorem 13.1 (No-Cloning from μ-Conservation (NoCloning.v))**.** *Let $I = x^2 + y^2 + z^2$ be the information content (purity). For a cloning operation with outputs of information I_1, I_2 and μ-cost μ:*

$$I_1 + I_2 \leq I + \mu$$

Theorem 13.2 (Approximate Cloning (NoCloning.v))**.** *For fidelities f_1, f_2: $f_1 + f_2 \leq 1 + \mu/I$. At zero cost: $f_1 + f_2 \leq 1$.*

Theorem 13.3 (No Deletion (NoCloning.v))**.** *Perfect deletion also requires $\mu \geq I$ (dual of no-cloning).*

Chapter 14

The Tsirelson Bound

Remark 14.1 (Epistemological Status: **(S)** algebraic, no physics axioms)**.** The active-corpus derivation lives in `coq/kernel/quantum/TsirelsonGeneral.v` (`tsirelson_from_row_bounds`) and `coq/kernel/category/AlgebraicCoherence.v` (`tsirelson_bound_tight`); both prove the bound algebraically without encoding $2\sqrt{2}$ into any definition. `TsirelsonGeneral.v` derives $S^2 \leq 8$ from pure algebra: a sum-of-squares identity proves each CHSH row contributes $\leq 2(e_1^2 + e_2^2)$, then Cauchy–Schwarz bounds the sum by $4\sum e_i^2 \leq 8$. `TsirelsonFromAlgebra.v` provides a self-contained algebraic derivation with achievability witness ($e = \pm 1/\sqrt{2}$, giving $S = 2\sqrt{2}$) and a verified rational bound ($\sqrt{8} < 5657/2000$). Zero admits in both files.

Theorem 14.1 (Tsirelson Upper Bound for the Zero-Marginal NPA Model (TsirelsonQuantumModel.v, TsirelsonFromAlgebra.v) — **(S)**)**.** *If a trace is quantum-realizable in the zero-marginal NPA model (i.e., is Thiele-honest / column-contractive), then $|S_{\mathrm{CHSH}}| \leq 2\sqrt{2}$.*

The algebraic derivation (`TsirelsonFromAlgebra.v`) proves $S^2 \leq 8$ for all algebraically coherent correlators satisfying the three column-contractivity conditions, using a sum-of-squares identity and psatz. No physics premises. Pure polynomial arithmetic.

Chapter 15

Witness Insight Taxonomy

Theorem 15.1 (CHSH Trial Is Tier 0 (kernel/WitnessInsightGeneral.v) — **(S)**). *`chsh_trial_is_tier0`: The CHSH_TRIAL instruction never touches either certification channel:*

- *`chsh_trial_not_cert_addr_setter`: CHSH_TRIAL does not change `csr_cert_addr`.*
- *`chsh_trial_preserves_vm_certified`: CHSH_TRIAL does not change `vm_certified`.*

Recording CHSH trial outcomes is provably free of structural cost. The counters go up; the cert channels are untouched. Cost: 0. Zero Admitted.

Theorem 15.2 (Three-Tier Observation Taxonomy (kernel/WitnessInsightGeneral.v) — **(S)**). *`witness_insight_complete_taxonomy`: Every observation the machine can make falls into exactly one of three tiers:*

Tier 0 *Raw observation (always free): Records data without activating any certification channel. CHSH_TRIAL is the canonical example. Cost: $\Delta\mu = 0$ is possible; no cert channel is touched.*

Tier 1 *Certified structural insight (always costs ≥ 1): Any instruction that activates `csr_cert_addr` or `vm_certified` is in this tier. Proven: `certified_witness_insight_nonfree` shows μ increases by at least 1.*

Tier 2 *Certified non-local witness insight (always costs ≥ 1): A certified execution whose WitnessCount statistics show $S > 2$. `nonlocal_witness_insight_nonfree`: getting certified with non-local evidence requires $\mu \geq 1$.*

`witness_insight_nonfree_general`: For any instruction in Tier 1 or Tier 2, μ increases by at least 1. The only free observations are Tier 0. Zero Admitted.

Chapter 16

Bell's Theorem in Morphism Coupling Language

Definition 16.1 (Setting-Outcome Coupling (kernel/CHSHCouplingBridge.v))**.** The `chsh_setting_outcome_c` maps a WitnessCount record to a coupling (list of ($nat \times nat$) pairs). Each pair records a (setting-pair-index, outcome-type) observation. A coupling is **separable** (functional) when it maps each setting to at most one outcome type; **non-separable** otherwise.

Theorem 16.1 (Locally Consistent Strategy Gives Separable Coupling (kernel/CHSHCouplingBridge.v) — **(S)**)**.** *`locally_consistent_gives_separable_coupling`: Any WitnessCount configuration produced by a locally deterministic strategy* $(a_0, a_1, b_0, b_1) \in \{0, 1\}^4$ *gives a separable (functional) coupling. Zero Admitted.*

Theorem 16.2 (Local Coupling Bound (kernel/CHSHCouplingBridge.v) — **(S)**)**.** *`locally_consistent_classi` Any locally factorizable morphism coupling satisfying `WCLocallyConsistent` has* $|S| \leq 2$*. Bell's theorem in the morphism coupling language: the separation between local and non-local correlations is not just a statistical fact but a structural property of the coupling predicate. Zero Admitted.*

Theorem 16.3 (CHSH Violation Rules Out Local Couplings (kernel/CHSHCouplingBridge.v) — **(S)**)**.** *`chsh_violation_rules_out_locally_factorizable_coupling`: If WitnessCount statistics satisfy* $S > 2$*, no locally factorizable morphism coupling is consistent with those counts. Contrapositively: non-local correlations cannot be produced by any local morphism coupling. The no-go is stated in the coupling language, not just the statistical layer. Zero Admitted.*

Remark 16.1. These results connect the categorical morphism layer directly to Bell nonlocality. A claim that two modules are entangled (via MORPH_ASSERT on a non-separable coupling) cannot be locally faked: the NoFI law charges at least 1 for the MORPH_ASSERT, and the coupling structure itself rules out any local hidden strategy that would produce $S > 2$ for free.

Chapter 17

Thiele Honesty and NPA Realizability: The Biconditional

This is a kernel-tier result: zero axioms, zero Admitted, pure polynomial arithmetic.

Definition 17.1 (Column Contractivity (Thiele Honesty))**.** A correlation tuple $(E_{00}, E_{01}, E_{10}, E_{11})$ is **Thiele-honest** (column-contractive) if:

$$1 - E_{00}^2 - E_{10}^2 \geq 0 \quad \text{(condition 1)} \tag{17.1}$$

$$1 - E_{01}^2 - E_{11}^2 \geq 0 \quad \text{(condition 2)} \tag{17.2}$$

$$(1 - E_{00}^2 - E_{10}^2)(1 - E_{01}^2 - E_{11}^2) \geq (E_{00}E_{01} + E_{10}E_{11})^2 \quad \text{(condition 3)} \tag{17.3}$$

These are exactly the PSD conditions on the 5×5 zero-marginal NPA moment matrix.

Theorem 17.1 (Thiele Honesty $\iff$ NPA Realizability (`coq/kernel/quantum/QuantumPartitionPSD.v`, corollary `column_contractive_iff_quantum_realizable`) — **(S)**)**.** *`column_contractive_iff_quantum_rea`*

$$\mathit{thiele\_honest}(\mathbf{E}) \iff \mathit{quantum\_realizable}(\mathbf{E})$$

where thiele_honest *means the three column-contractivity conditions hold, and* quantum_realizable *means the 5×5 zero-marginal NPA moment matrix is positive semidefinite.*

Forward direction *(`zero_marginal_npa_column_contractive_implies_psd` in `coq/kernel/nfi/MuLed` column-contractive ⇒ NPA PSD. Proved by constructing the Cholesky decomposition of the 5×5 matrix from the three conditions.*

Reverse direction *(`npa_psd_implies_column_contractive` in `coq/kernel/quantum/QuantumPartitio` NPA PSD ⇒ column-contractive. Proved by specializing the PSD condition at the vector $(0, -(E_{00}y_0 + E_{01}y_1), -(E_{10}y_0 + E_{11}y_1), y_0, y_1)$, obtaining a non-negative quadratic in y_0, y_1, then closing via `quadratic_nonneg_discriminant`.*

This is a zero-axiom biconditional. No physics assumptions. Pure polynomial arithmetic. The zero-marginal NPA framework is the orthogonal-observables slice of the quantum elliptope ($\rho_{AA} = \rho_{BB} = 0$), not the full elliptope.

Key consequences:

- *PR box ($E_{00} = E_{01} = E_{10} = 1, E_{11} = -1$) fails condition 1 immediately ($1-1-1 = -1 < 0$). Direct arithmetic; no separate theorem needed.*

- *`state_column_contractive_implies_tsirelson` (`coq/kernel/nfi/MuLedgerQuantumBridge.v`) directly proves Thiele-honest ⇒ $|S|^2 \leq 8$ (equivalently $|S| \leq 2\sqrt{2}$); its proof composes `state_column_contractive_implies_quantum_gram` with `quantum_realizable_implies_tsirelson_` The Cauchy–Schwarz inner step (PSD ⇒ row bounds ⇒ $|S|^2 \leq 8$) is `tsirelson_from_row_bounds` in `coq/kernel/quantum/TsirelsonGeneral.v`. The contrapositive ($|S| > 2\sqrt{2}$ ⇒ not Thiele-honest) follows immediately. The converse is false and is not claimed.*

- *Turing-classical correlations* $(1, 0, 1, 0)$ *satisfy* $|S| = 2$ *but fail condition 1:* $1-1-1 = -1 < 0$. *A Turing machine can output this correlation; the Thiele Machine rejects it as dishonest. Direct arithmetic from* `column_contractive_iff_quantum_realizable`.

Zero Admitted. Zero named hypotheses.

Chapter 18

Emergent Spacetime

18.1 The μ-Metric

Definition 18.1 (μ-Distance (MuGeometry.v))**.** The Coq formalization defines `mu_distance` as the absolute difference of μ-totals:

$$d_\mu(S_1, S_2) = |\mu_{\text{total}}(S_2) - \mu_{\text{total}}(S_1)|$$

This is a one-dimensional projection, not a minimum over paths.

Theorem 18.1 (Pseudometric Properties (MuGeometry.v))**.** *d_μ satisfies non-negativity, reflexivity ($d(S,S) = 0$), symmetry, and triangle inequality, all inherited from absolute value on $\mathbb{Z}$. Note: d_μ is a **pseudometric**, not a metric: $d(S_1, S_2) = 0$ does* not *imply $S_1 = S_2$ (many distinct states share the same μ-total). The "geometry" is a 1D projection onto the μ counter.*

18.2 Causal Cones

Definition 18.2 (μ-Reachability Cone (SpacetimeEmergence.v))**.** $C^+(S) = \{S' \mid \exists \tau.\ S \xrightarrow{\tau} S' \wedge \text{cost}(\tau) \leq \text{Budget}\}$.

This is a structural analogy to causal cones: the set of reachable states given a finite μ-budget plays a role analogous to a light cone. The analogy should not be over-read: there is no Lorentz invariance, no metric signature, and no boost transformations in the Coq formalization.

18.3 No-Signaling

Theorem 18.2 (Observational Locality (SpacetimeEmergence.v))**.** *If module M_B is not in the target set of any instruction in a trace, then M_B's observable region is unchanged after execution (`exec_trace_no_signaling_outside_cone`).*

Remark 18.1. This is a **data isolation** property: operations targeting module A in the partition graph do not affect the lookup of module B. The proof is structural (list operations on disjoint keys), non-trivial mainly for `instr_psplit`/`instr_pmerge` which restructure the graph. The label "no-signaling" is an analogy. There is no spatial metric or speed limit between modules.

18.4 Discrete Geometry: The Gauss-Bonnet Result

The partition graph is a discrete geometric object. One unconditional result falls out, requiring no physics axioms.

Theorem 18.3 (Discrete Angle-Defect Identity (Planar Triangulations) (`coq/kernel/curvature/DiscreteGaus` — **(S)**)**.** *For any* `well_formed_triangulated` *partition graph g in the equilateral angle model (each triangle corner contributes $\pi/3$):*

$$\text{angle\_defect}(g) = 5\pi \cdot \chi(g)$$

where $\chi(g) = V - E + F$ is the Euler characteristic (vertices minus edges plus faces), and $\text{angle\_defect}(g)$ *is the sum over all vertices of 2π minus the total incident angle. This is a fully proven combinatorics result about the discrete triangulation model. It is not the continuum Gauss-Bonnet theorem and it is not a derivation of angles from μ-cost. Zero Admitted.*

Remark 18.2 (How this differs from classical Gauss-Bonnet)**. The factor 5π is not a typo and not the classical Gauss-Bonnet constant.** For a closed triangulated 2-manifold (like the surface of a tetrahedron), the standard derivation gives $D = 2\pi\chi$: every interior edge is shared by exactly two faces, so $3F = 2E$, and substituting that into $D = 2\pi V - \pi F$ together with $\chi = V - E + F$ yields $D = 2\pi\chi$. The Coq predicate `well_formed_triangulated` does *not* include $3F = 2E$. It instead requires `satisfies_boundary_euler_relation` ($B = 3\chi$, where B is the boundary edge count) together with $E = I + B$ and $3F = 2I + B$. From these the file derives the combinatorial identity $3V = 5E - 6F$, which in turn forces $D = 5\pi\chi$.

That identity holds for **planar triangulations with boundary** (the canonical example: a planar disk with $V = 7$, $E = 15$, $F = 9$, $\chi = 1$, satisfying $3V = 21 = 5E - 6F$ and $D = 5\pi$). It **fails** for closed surfaces (the tetrahedron has $V = 4$, $E = 6$, $F = 4$, but $3V = 12 \neq 6 = 5E - 6F$, so it is excluded from the predicate). The theorem is therefore not a discrete Gauss-Bonnet theorem in the classical sense — it is a different combinatorial identity for a different family of complexes. The two coexist: classical Gauss-Bonnet at $2\pi\chi$ for closed 2-manifolds, the file's identity at $5\pi\chi$ for planar disks satisfying its well-formedness predicate.

This is the full extent of the topology-curvature identity in the project.

18.4.1 Cross-link in MuGravity Vocabulary

Theorem 18.4 (Graph-Laplacian Sum-Zero Identity (`coq/kernel/frontier/F3_MuLaplacianSum.v`) — **(S)**)**.** *For every* `VMState` *s:*

$$\sum_{m \in \text{pg\_modules}(\text{vm\_graph}\, s)} \text{mu\_laplacian}(s, m) = 0.$$

Proof by edge-pair antisymmetry. `nat_list_disjoint` *is symmetric, hence* `modules_adjacent_by_region` *is symmetric, hence* `edge_weight` *is symmetric;* `mu_gradient` *is antisymmetric by definition; the double sum cancels under the Fubini swap. No well-formedness, no calibration. Standalone graph-theoretic fact.*

Theorem 18.5 (F3 Narrow Cross-link (`coq/kernel/frontier/F3_PartitionTopologyCrossLink.v`) — **(S)**)**.** *For every* `VMState` *s at universal local calibration ($\forall m$,* `calibration_residual`$(s, m) = 0$*):*

$$\sum_{m \in \text{pg\_modules}(\text{vm\_graph}\, s)} \text{geometric\_angle\_defect}(s, m) = 0.$$

The proof composes two independent kernel facts: `calibration_residual_zero_iff` *(geometry $\leftrightarrow$ μ-ledger at every module) and* `total_mu_laplacian_zero` *(Theorem 18.4). Both ingredients are load-bearing — drop counter-instances* `F3_drop_calibration_breaks_prediction` *and* `F3_drop_sum_zero_breaks_prediction` *retained.*

Remark 18.3 (Why this is not the topology version)**.** An earlier form of this cross-link concluded $\sum_m \text{mu\_laplacian} = 5 \cdot \chi$ by composing the discrete angle-defect identity (Theorem 18.4) with a sum-form calibration. That composition silently identifies two distinct angle-defect quantities: `DiscreteGaussBonnet.angle_defect` (degree $\times\ \pi/3$, equilateral) and `MuGravity.geometric_angle_defect` (distance-weighted, $\pi \cdot d_{bc}/(1+d_{ab}+d_{ac}+d_{bc})$). They are not equal in general: `strong_bridge_counterexample` (in the same file) formalises a degree-6 vertex at uniform inter-module distance $d \geq 1$ where the Gauss-Bonnet defect is 0 and the MuGravity defect is $2\pi/(1+3d) \neq 0$. The narrow form drops the topology invariant. Discrete Gauss-Bonnet remains a kernel theorem; it is simply not load-bearing for this cross-link.

Non-vacuity is open. Symbolic search across small actually-triangulated graphs (K_4, $K_4 - e$, octahedron, prism, K_5, B_3, K_6, K_7) up to $N = 7$ returns no integer-mass calibrated solution; structural analysis identifies three failure modes (insufficient triangles, excess triangles, asymptotic-only target reachability). `module_triangles` was tightened in `MuGravity.v` to require pairwise-adjacent neighbors (an actual graph triangle), eliminating tree-shaped pseudo-witnesses.

18.5 Four-Dimensional Spacetime Geometry (Reference Only, Not a Main Claim)

Remark 18.4 (Scope Warning)**.** The following subsections define discrete differential geometry objects (simplicial complex, metric tensor, Christoffel symbols, Riemann tensor, Einstein tensor) over partition graphs and prove results about the vacuum case $G_{\mu\nu} = 0 = T_{\mu\nu}$. This is a $0 = 0$ result: both sides are zero in the vacuum because mass is zero everywhere. It is **not** a derivation of Einstein's field equations from computation. The gravitational constant $G := 1/(8\pi)$ is a unit choice (a definition), not a derived quantity. No physical claim is being made. The geometry section's main result is the Discrete Gauss-Bonnet theorem (Section 18.4). Everything in this section is marked reference-only and does not appear in the primary claim ledger.

The previous sections treated spacetime as a 1D projection via the μ-metric. The next subsections extend to full 4D geometry as a reference exploration. None of the results here are part of the core formal claims of the project.

18.5.1 4D Simplicial Complex

Definition 18.3 (4D Simplicial Complex (FourDSimplicialComplex.v))**.** A **4D simplicial complex** $\mathcal{K}$ is a discrete approximation to 4D spacetime consisting of:

- **Vertices** V: Computational modules (ModuleID)
- **Edges** $E \subseteq V \times V$: Connections between modules
- **Simplices**: 2-simplices (triangles), 3-simplices (tetrahedra), 4-simplices (4D cells)

The complex satisfies:

- **Consistency**: Every face of a simplex is also in $\mathcal{K}$
- **Well-formedness**: Dimension 4 with locally Euclidean structure

18.5.2 Metric Tensor from μ-Costs

Definition 18.4 (Computational Metric (MetricFromMuCosts.v, RiemannTensor4D.v))**.** The metric tensor at vertex v with respect to vertices w_1, w_2 is:

$$g_{\mu\nu}(v; w_1, w_2) = \begin{cases} \ell(v, w_1, w_2) & \text{if } \mu = \nu \\ 0 & \text{if } \mu \neq \nu \end{cases} \tag{18.1}$$

where the edge length function is:

$$\ell(s, v_1, v_2) = 2 \cdot \text{mass}(v_1) \quad \text{if } v_1 = v_2 \tag{18.2}$$

and mass is derived from structural complexity: $\text{mass}(v) = |\text{module\_structural\_mass}(s, v)|$.

Remark 18.5. This metric is **position-dependent** for non-uniform mass distributions:

- **Uniform mass** m: $g_{\mu\mu}(v; w, w) = 2m$ (constant) $\implies$ flat spacetime
- **Non-uniform mass**: $g_{\mu\mu}(v; w, w)$ varies with $w \implies$ curved spacetime

In this model, mass distributions create position-dependent geometry: uniform mass gives flat spacetime, non-uniform mass gives curved spacetime.

18.5.3 Discrete Differential Geometry

Definition 18.5 (Discrete Derivative (RiemannTensor4D.v))**.** For a scalar field $f : V \to \mathbb{R}$ and direction μ:

$$\partial_\mu f(v) = \frac{1}{|N_\mu(v)|} \sum_{w \in N_\mu(v)} (f(w) - f(v)) \tag{18.3}$$

where $N_\mu(v)$ is the set of neighbors of v in the μ-direction.

Definition 18.6 (Christoffel Symbols (RiemannTensor4D.v))**.** The connection coefficients (Christoffel symbols of the second kind) are:

$$\Gamma^\rho_{\mu\nu}(v) = \frac{1}{2} \sum_\sigma g^{\rho\sigma}(v) \Big(\partial_\mu g_{\nu\sigma}(v) + \partial_\nu g_{\mu\sigma}(v) - \partial_\sigma g_{\mu\nu}(v) \Big) \tag{18.4}$$

where $g^{\rho\sigma}$ is the inverse metric (diagonal in this construction).

Lemma 18.6 (Flat Spacetime Has Zero Connection (EinsteinEquations4D.v) — **(S)**)**.** *For uniform mass distribution:*

$$\forall w,\ mass(w) = m \quad \implies \quad \Gamma^\rho_{\mu\nu}(v) = 0 \tag{18.5}$$

Proof. Uniform mass $\implies$ position-independent metric $\implies \partial_\mu g_{\nu\sigma} = 0 \implies \Gamma^\rho_{\mu\nu} = 0$. □

18.5.4 Curvature Tensors

Definition 18.7 (Riemann Curvature Tensor (RiemannTensor4D.v))**.** The Riemann curvature tensor measures the failure of parallel transport:

$$\begin{aligned} R^\rho_{\sigma\mu\nu}(v) = \partial_\mu \Gamma^\rho_{\nu\sigma}(v) - \partial_\nu \Gamma^\rho_{\mu\sigma}(v) \\ + \sum_\lambda \Big(\Gamma^\rho_{\mu\lambda}(v) \Gamma^\lambda_{\nu\sigma}(v) - \Gamma^\rho_{\nu\lambda}(v) \Gamma^\lambda_{\mu\sigma}(v) \Big) \end{aligned} \tag{18.6}$$

Definition 18.8 (Ricci Tensor (RiemannTensor4D.v))**.** Contraction of the Riemann tensor:

$$R_{\mu\nu}(v) = \sum_\rho R^\rho_{\mu\rho\nu}(v) \tag{18.7}$$

Definition 18.9 (Ricci Scalar (RiemannTensor4D.v))**.** Complete contraction:

$$R(v) = \sum_{\mu,\nu} g^{\mu\nu}(v) R_{\mu\nu}(v) \tag{18.8}$$

Definition 18.10 (Einstein Tensor (RiemannTensor4D.v))**.** The Einstein tensor encodes spacetime curvature:

$$G_{\mu\nu}(v) = R_{\mu\nu}(v) - \frac{1}{2} g_{\mu\nu}(v) R(v) \tag{18.9}$$

Lemma 18.7 (Flat Spacetime Has Zero Curvature (EinsteinEquations4D.v) — **(S)**)**.** *For uniform mass:*

$$\forall w,\ mass(w) = m \quad \implies \quad R^\rho_{\sigma\mu\nu}(v) = 0 \quad \implies \quad G_{\mu\nu}(v) = 0 \tag{18.10}$$

18.5.5 Stress-Energy Tensor

Definition 18.11 (Energy Density (EinsteinEquations4D.v))**.** The energy density is the computational mass:

$$T_{00}(v) = \text{mass}(v) \tag{18.11}$$

Definition 18.12 (Momentum Density (EinsteinEquations4D.v))**.** Spatial derivatives of energy:

$$T_{0i}(v) = T_{i0}(v) = \partial_i T_{00}(v) \tag{18.12}$$

Definition 18.13 (Stress Components (EinsteinEquations4D.v))**.** Spatial stress tensor:

$$T_{ij}(v) = \begin{cases} \text{mass}(v) & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \tag{18.13}$$

Definition 18.14 (Stress-Energy Tensor (EinsteinEquations4D.v))**.** Combining all components:

$$T_{\mu\nu}(s, \mathcal{K}, v) = \begin{cases} T_{00}(v) & \text{if } \mu = \nu = 0 \\ T_{0i}(v) & \text{if } \mu = 0, \nu = i \text{ or } \mu = i, \nu = 0 \\ T_{ij}(v) & \text{if } \mu = i, \nu = j \\ 0 & \text{otherwise} \end{cases} \tag{18.14}$$

18.5.6 Einstein's Field Equations

Definition 18.15 (Gravitational Constant (EinsteinEquations4D.v))**.** In computational units where the fundamental scale is set by μ-costs:

$$G = \frac{1}{8\pi} \tag{18.15}$$

This makes $8\pi G = 1$, normalizing the coupling in Einstein's equations.

Lemma 18.8 (Curvature from μ-Gradients (EinsteinEquations4D.v) — **(S)**)**.** *For uniform mass distribution:*

$$\forall w,\ mass(w) = m \quad \Longrightarrow \quad G_{\mu\nu}(v) = 0 \tag{18.16}$$

Proof. The proof chain is:

1. Uniform mass $\Longrightarrow$ position-independent metric
2. Position-independent metric $\Longrightarrow$ zero discrete derivatives $\partial_\mu g_{\nu\sigma} = 0$
3. Zero metric derivatives $\Longrightarrow$ zero Christoffel symbols $\Gamma^\rho_{\mu\nu} = 0$
4. Zero Christoffel $\Longrightarrow$ zero Riemann tensor $R^\rho_{\sigma\mu\nu} = 0$
5. Zero Riemann $\Longrightarrow$ zero Ricci tensor and scalar
6. Zero Ricci $\Longrightarrow$ zero Einstein tensor $G_{\mu\nu} = 0$

All steps are proven constructively in `EinsteinEquations4D.v`. □

Lemma 18.9 (Vacuum Implies Zero Stress-Energy (EinsteinEquations4D.v) — **(S)**)**.** *For vacuum states (zero mass everywhere):*

$$\forall w,\ mass(w) = 0 \quad \Longrightarrow \quad T_{\mu\nu}(v) = 0 \tag{18.17}$$

Proof. Direct computation:

- $T_{00} = \text{mass} = 0$
- $T_{0i} = \partial_i(\text{mass}) = \partial_i(0) = 0$
- $T_{ij} = \delta_{ij} \cdot \text{mass} = \delta_{ij} \cdot 0 = 0$

□

Theorem 18.10 (Vacuum Case: Both Sides Are Zero (`coq/kernel/curvature/EinsteinEquations4D.v`) — **(R)**)**.** *For any VM state s, 4D simplicial complex $\mathcal{K}$, spacetime indices μ, ν, and vertex v:*

$$\forall w,\ \mathit{mass}(w) = 0 \quad \Longrightarrow \quad G_{\mu\nu}(s, \mathcal{K}, v) = 8\pi G\, T_{\mu\nu}(s, \mathcal{K}, v) \tag{18.18}$$

Honest status: this is $0 = 0$**.** *Both sides equal zero in vacuum because all masses vanish. The gravitational constant $G := 1/(8\pi)$ is a definition (unit choice), not a derived quantity. This is* ***not*** *a derivation of Einstein's field equations. It is a consistency check that the discrete geometric objects reduce correctly when mass is everywhere zero. The non-vacuum case is not proven. This theorem does not appear in the primary claim ledger of the project.*

Proof. **Step 1**: Expand the right-hand side using $G = 1/(8\pi)$:

$$8\pi G\, T_{\mu\nu} = 8\pi \cdot \frac{1}{8\pi} \cdot T_{\mu\nu} = T_{\mu\nu} \tag{18.19}$$

Step 2: Apply Lemma 18.8 (curvature from μ-gradients):
Vacuum (mass $= 0$ everywhere) is a special case of uniform mass distribution, so:

$$G_{\mu\nu}(v) = 0 \tag{18.20}$$

Step 3: Apply Lemma 18.9 (stress-energy conservation):
For vacuum:

$$T_{\mu\nu}(v) = 0 \tag{18.21}$$

Conclusion: Both sides equal zero:

$$0 = 0 \quad \checkmark \tag{18.22}$$

The complete proof is formalized in `coq/kernel/curvature/EinsteinEquations4D.v` with:

- No project-local axioms
- Zero admitted lemmas
- Zero classical logic imports
- Verified by automated proof auditing (Inquisitor: 0 HIGH findings)

□

Remark 18.6 (Vacuum vs. Non-Vacuum)**.** Theorem 18.10 requires the vacuum hypothesis (mass $= 0$ everywhere). For non-vacuum states with arbitrary mass distributions, the equation $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ only holds when the distribution satisfies discrete Bianchi identities (conservation laws). This is analogous to classical general relativity, where Einstein's equations are *field equations* that constrain which configurations are physically possible, not universal identities satisfied by all states.

The vacuum case is sufficient to demonstrate that spacetime curvature emerges from computational dynamics. The non-vacuum case requires discrete differential geometry beyond the current formalization (specifically, discrete covariant derivatives and discrete Bianchi identities).

Remark 18.7 (Epistemological Status: **(R)** Consistency Relation)**.** This is a **(R)** result: the vacuum theorem reduces to $0 = 0$ (both sides vanish when all masses are zero; the gravitational constant is a definition). It establishes that the discrete geometric objects are internally consistent in the zero-mass case. It does *not* establish that spacetime geometry emerges from computation in any physically meaningful sense; that would require the A_QM named hypothesis and is an open empirical question. The non-vacuum case (the full Einstein field equations for arbitrary mass distributions) is not proved and is not a claim of this project.

Remark 18.8. The discrete geometric objects (metric, curvature, Einstein tensor) are defined from computational primitives. In the vacuum case (all masses zero) they all reduce to zero. This is a consistency check, not a derivation of spacetime emergence. The claim that spacetime geometry is an emergent property of computation would require more: a non-trivial non-vacuum case, a connection to physical observations, and the named A_QM bridge. None of those are present here.

18.5.7 Empirical Validation

The theoretical results are validated numerically:

Theorem 18.11 (Numerical Verification (tests/test_*) — Empirical)**.**

1. ***Flat spacetime*** *(uniform mass $m = 1$):*
   - $\max |\Gamma^\rho_{\mu\nu}| < 10^{-10}$ ✓
   - $\max |R^\rho_{\sigma\mu\nu}| < 10^{-10}$ ✓
   - $\max |G_{\mu\nu}| < 10^{-10}$ ✓
2. ***Curved spacetime*** *(non-uniform mass $m \in [0.5, 1.5]$):*
   - $\max |\Gamma^\rho_{\mu\nu}| \approx 3.0$ ✓ *(curvature detected)*
   - $\max |R^\rho_{\sigma\mu\nu}| > 0$ ✓
3. ***Vacuum Einstein equations*** *(mass $= 0$):*
   - $|G_{\mu\nu}| < 10^{-10}$ ✓
   - $|T_{\mu\nu}| < 10^{-10}$ ✓
   - $|G_{\mu\nu} - 8\pi G T_{\mu\nu}| < 10^{-10}$ ✓

These tests confirm that the discrete differential geometry correctly implements the theoretical predictions.

18.5.8 Affine Metric-Scaled Operator: Non-Uniform Boundary Results

The first-neighbor discrete operator refutes the non-uniform diagonal extension of the EFE. A signed affine metric-scaled redesign closes the gap at vertex 1 and fully characterizes the outer vertices.

Theorem 18.12 (Affine Off-Diagonal Ricci Zero (kernel/AffineEFEClosure.v) — **(S)**)**.** *affine_off_diagonal_r* *Under the affine metric-scaled symmetric operator, with the non-uniform isotropic boundary 4-simplex metric, off-diagonal Ricci vanishes at vertex 1: $R_{\mu\nu}(v_1) = 0$ for $\mu \neq \nu$. Zero Admitted. Zero Section Variables.*

Theorem 18.13 (Full Tensor EFE at Vertex 1 (kernel/AffineEFEClosure.v) — **(S)**). `affine_full_tensor_efe:` *Under* `module_structural_mass s 1 = 1` *(unit structural mass at vertex 1), the full tensor Einstein field equation holds:*

$$G_{\mu\nu}(v_1) = 3 \cdot T_{\mu\nu}(v_1)$$

for all $\mu, \nu \in \{0, 1, 2, 3\}$*. Zero Admitted.*

Theorem 18.14 (Outer Vertex Characterization (kernel/AffineEFEClosure.v) — **(S)**). *At the outer vertices* $v \in \{0, 2, 3, 4\}$ *of the boundary 4-simplex, the affine operator produces:*

- `affine_ricci_offdiag_outer`*: Off-diagonal Ricci* $= 3/32$ *(non-zero).*
- `affine_ricci_diag_outer`*: Diagonal Ricci* $= -45/32$*.*
- `affine_ricci_scalar_outer`*: Ricci scalar* $= -45/8$*.*
- `affine_einstein_outer`*: Einstein tensor* $= 45/32$ *(diagonal) and* $3/32$ *(off-diagonal).*
- `affine_efe_fails_outer_offdiag`*: The diagonal EFE form* $G = 3T$ *fails at outer vertices because* G *has non-zero off-diagonal components.*

The affine factor $8 - 5g_{00}$ *is specifically tuned to the center vertex geometry; at outer vertices it produces a non-trivial off-diagonal Einstein tensor. Zero Admitted.*

Remark 18.9 (What Remains Open)**.** The affine operator closes vertex 1 and characterizes the outer vertices. The remaining open question is whether the affine factor $8 - 5g_{00}$ is the correct physical discretization. That requires connecting the discrete operator to a physical substrate — not a Coq question, an empirical one.

Chapter 19

Information Causality

Theorem 19.1 (IC–μ Equivalence (InformationCausality.v) — **(R)**)**.** *The file defines two record types (`ICScenario` and `MuScenario`) with opaque `Prop` fields, then proves they are equivalent* given *a hypothesis (`ic_mu_equivalent`) that literally states the equivalence. The proof is `tauto`. No mutual information, probability distributions, or CHSH values appear anywhere in the formalization.*

Remark 19.1. The intended statement — that Bob's total information about Alice's data is bounded by the channel's μ-capacity — is a reasonable physical claim but is **not formalized** in the Coq file. The current file is a placeholder that assumes the conclusion as a hypothesis and re-derives it. A future version would need to define entropy, mutual information, and prove the bound from the VM semantics.

Chapter 20

Thermodynamic Bridge

20.1 Landauer's Principle

Theorem 20.1 (Landauer Derived (LandauerDerived.v))**.**

1. *`erasure_irreversible`: Erasing* ≥ 1 *bit implies fan-in* $= 2^k > 1$ *(arithmetic:* $2^k > 1$ *for* $k \geq 1$*).*
2. *`erasure_decreases_entropy`: Erasure decreases system entropy (*$\Delta S < 0$*) — this is just* $\text{output} < \text{input} \Rightarrow \text{output} - \text{input} < 0$*.*
3. *`landauer_information_bound`: For any physical erasure, environment entropy increase* $\geq$ *bits erased.*
4. *`erasure_additive`: Sequential erasures compose additively.*

Caveat on item 3*: The `PhysicalErasure` record has `second_law_satisfied` as a* ***field****, not a derived property. The theorem `landauer_information_bound` simply extracts this field, making the bound an* input assumption *rather than a derivation. The physical energy bound* $Q \geq k_B T \ln 2 \cdot n$ *is* ***explicitly not proven*** *(the file itself notes the conversion factor is not formalized). Also, the `mu` in this file is a standalone synonym for `bits_erased`—it has no formal connection to `vm_mu` in `VMState.v`.*

20.2 A2 from Landauer (Factored Form)

Theorem 20.2 (F1 Strong Form: A2 via Physical Landauer (`coq/kernel/frontier/F1_StrongForm.v`) — **(C)**)**.** *Given an abstract dissipation function* $\phi : \text{vm\_instruction} \to \mathbb{N}$ *(interpreted as dissipation per step in Landauer quanta), if*

1. ***Landauer's principle*** *holds in dissipation vocabulary — every class-collapsing step on any bool macro-property has* $\phi(i) \geq 1$*, and*
2. *the named bridge `cost_dissipation_calibrated` holds —* $\text{instruction\_cost}(i) \geq \phi(i)$ *for every* i*,*

then on every cert-flip step $\text{instruction\_cost}(i) \geq 1$*. The Landauer premise mentions only class-collapse and dissipation; `instruction_cost` appears only in the calibration. Both ingredients are independent and load-bearing: dropping either leaves the conclusion unprovable.*

The factorisation is what F1 buys. The strict reading of "derive A2 from physical reversibility, with no cost-language anywhere in the premise" is unattainable: A2's conclusion is itself a cost statement, so any unit-conversion bridge necessarily mentions cost (the same shape Newton's laws + Newtons take). The factored form isolates the cost interface to a single named calibration, leaving the Landauer principle in dissipation vocabulary alone.

Chapter 21

Self-Reference and the Thiele Manifold

21.1 Self-Reference Escalation

Theorem 21.1 (Gödel-Style Escalation (SelfReference.v)). *Any self-referential system S requires a meta-system* Meta *with strictly more dimensions:*

$$\text{self\_ref}(S) \implies \exists\,\text{Meta} : \dim(\text{Meta}) > \dim(S)$$

The meta-system inherits self-reference, creating an infinite escalation.

21.2 The Thiele Manifold

Definition 21.1 (Thiele Manifold (ThieleManifold.v)). An infinite tower of systems $\{L_n\}_{n\in\mathbb{N}}$ with:

- $\dim(L_n) = 4 + n$ (level n has dimension $4 + n$).
- $\dim(L_{n+1}) > \dim(L_n)$ (strict enrichment).
- Each level reasons about the level below.
- Self-reference at level n is answered by level $n + 1$.

Theorem 21.2 (Projection to Spacetime (ThieleManifold.v)). *The projection π_4 collapses the tower to dimension 4 (spacetime). For $n > 0$:*

- *π_4 is lossy:* $\dim(L_n) > 4$.
- *μ-cost of projection* $= \dim(L_n) - 4 = n > 0$.

Spacetime is the "shadow" of the full manifold.

Chapter 22

Three-Layer Isomorphism

Theorem 22.1 (Observable Comparison Contract (ThreeLayerIsomorphism.v))**.** *The active Coq theorem packages the* observable-state contract *used for cross-layer comparison: if the chosen Python and RTL decoders satisfy the same projection contract as the Coq layer, then the decoded observable states agree on that projection for a shared trace τ.*

$$\mathrm{decode}_{\mathrm{Python}}(\tau) = \mathrm{decode}_{\mathrm{Coq}}(\tau) \land \mathrm{decode}_{\mathrm{RTL}}(\tau) = \mathrm{decode}_{\mathrm{Coq}}(\tau) \Longrightarrow \mathrm{decode}_{\mathrm{Python}}(\tau) = \mathrm{decode}_{\mathrm{RTL}}(\tau)$$

This is a contract theorem over explicit observable projections, not an unconditional claim that every current Python and RTL artifact is universally equal to Coq in all CI environments.

22.1 RTL Coverage: Zero Gaps

Theorem 22.2 (RTL Gap Registry Empty (kami_hw/RTLGapRegistry.v) — **(S)**)**.** *`rtl_gap_count`: The RTL gap registry is empty. There are zero unresolved coverage gaps between the Kami hardware specification and the Coq kernel semantics:*

$$\mathit{rtl\_gap\_count} = 0$$

`rtl_coverage_partition`: The 47 synth-realized opcodes (of the 51-opcode ISA) partition as:

$$37\ (\textit{unconditional Qed}) + 10\ (\textit{Qed under inductive invariants } \mathtt{coupling\_wf},\ \mathtt{morph\_table\_wf}) + 0\ (\textit{gaps}) = 47$$

This is a Coq theorem (`rtl_coverage_partition` in `coq/kami_hw/RTLGapRegistry.v`), not a comment. The partition equation is machine-checked. Zero Admitted.

Remark 22.1. The official partition (`rtl_coverage_partition` in `coq/kami_hw/RTLGapRegistry.v`) is $37 + 10 + 0 = 47$. The 37 unconditional proofs are: 31 from the SupportedOpcode group, including `CHSH_LASSERT` (`coq/kami_hw/EmbedStep.v: embed_step_compute`) plus CALL, RET, CHSH_TRIAL (`coq/kami_hw/EmbedStep_WF.v`) and TENSOR_SET, TENSOR_GET, LASSERT (`coq/kami_hw/GraphReconstructionBridge.v`). The remaining 10 (PNEW, PSPLIT, PMERGE, MORPH, MORPH_ID, MORPH_DELETE, MORPH_ASSERT, MORPH_GET, COMPOSE, MORPH_TENSOR) carry Qed proofs under the joint structural invariant `morph_table_wf` $\land$ `coupling_wf` $\land$ `coupling_desc_safe`. Each component is preserved by every `kami_step`: `morph_table_wf_kami_step_preserved`, `coupling_desc_safe_kami_step_preserved`, and `coupling_wf_ka` The last takes `coupling_desc_safe` as a side hypothesis (in addition to `coupling_wf`), which is why the conjunction is the actual inductive invariant. So the precondition is discharged for any state reachable by machine execution. PNEW/PSPLIT/PMERGE additionally need `pt_well_formed` and arithmetic side conditions that are likewise inductive.

22.2 Named Trust Boundaries

The project has exactly two named trust boundaries. They are not Admitted proofs. They are explicit, auditable limits of the formal claim.

`bsc_kami_compilation_trusted` (Layer 2, Verilog semantics gap). The Bluespec SystemVerilog compiler generates Verilog from the Kami hardware specification. The correctness of the BSC compilation step — that the generated Verilog implements exactly the Kami spec — is trusted, not proved in Coq. This is documented in `coq/kernel/hardware_bridge/VerilogRTLCorrespon` (Section 2b, "BSC Compilation Trust Boundary") and the matching `Definition bsc_kami_compilation_` The Section Variable `rtl_step_correct` states that the synthesised RTL implements the Kami step function; this is the residual hypothesis that every theorem in the section quantifies over. The boundary is empirically backed by the cosim suite (`tests/test_verilog_cosim.py`, 38 tests, including `test_all_47_opcodes_in_cosim` which checks that every opcode appears) and the parametrized cross-layer fuzz suites (`tests/test_fuzz_random_programs.py` and `tests/test_cross_layer_adversarial_fuzz.py`).

`A_QM` (physical quantum axiom). Whether a physical Thiele Machine can produce CHSH-violating statistics by exploiting quantum entanglement in hardware is an open empirical question. The `column_contractive_iff_quantum_realizable` biconditional (Theorem 17.1) requires no physical axiom — it is pure algebra. The physical quantum axiom `A_QM` (declared as a `Context` parameter in `coq/PhysicsConditionalClosure.v`) is the separate named hypothesis that quantum CHSH experiments produce PSD NPA matrices. It does not appear in the kernel-tier proof of the biconditional; it is the bridge used in conditional theorems such as `master_tsirelson_conditional`.

Chapter 23

Categorical Structure

23.1 The Morphism Layer

The active implementation adds a concrete category directly to the machine state. This is not an abstract analogy — it is an executable layer of 7 opcodes (0x27–0x2D) with formally verified laws.

23.1.1 Objects and Morphisms

The partition graph G now carries two maps:

- `pg_modules` : ModuleID $\rightharpoonup$ ModuleState — objects (partition modules)
- `pg_morphisms` : MorphismID $\rightharpoonup$ MorphismState — arrows

A MorphismState bundles: source module, target module, coupling data (list of $(nat \times nat)$ pairs plus a label), and an identity flag.

23.1.2 The Seven Categorical Opcodes

- `MORPH`$(dst, src, tgt, idx, \Delta\mu)$: create morphism $src \to tgt$, write morphism ID to `regs`$[dst]$. Costs $\Delta\mu$.
- `COMPOSE`$(dst, f, g, \Delta\mu)$: compose morphisms $f; g$ when f.target $= g$.source, write new ID to `regs`$[dst]$. Costs $\Delta\mu$. Errors if endpoints mismatch.
- `MORPH_ID`$(dst, m, \Delta\mu)$: create identity morphism id_m for module m (diagonal relation). Costs $\Delta\mu$.
- `MORPH_DELETE`$(f, \Delta\mu)$: remove morphism f. Cascades on module deletion via `graph_cascade_delete_mor`
- `MORPH_ASSERT`$(f, prop, cert, \Delta\mu)$: cert-setter. Costs $S(\Delta\mu) \geq 1$ unconditionally — even for failed attempts (morphism does not exist). This is the No Free Insight law applied to relational claims.
- `MORPH_TENSOR`$(dst, f, g, \Delta\mu)$: tensor product $f \otimes g$. Requires that the union modules $\mathrm{src}(f) \cup \mathrm{src}(g)$ and $\mathrm{tgt}(f) \cup \mathrm{tgt}(g)$ already exist in the graph.
- `MORPH_GET`$(dst, f, sel, \Delta\mu)$: read field from morphism f: selector 0 = source, 1 = target, 2 = coupling length, 3 = is_identity.

23.1.3 Category Laws

Theorem 23.1 (Relational Composition is Associative (kernel/CategoryLaws.v) — **(S)**)**.** *For any three morphisms $f : A \to B$, $g : B \to C$, $h : C \to D$ with composable endpoints:*

$$(f; g); h = f; (g; h)$$

in the relational sense (coupling pairs of the composite). Zero Admitted.

Theorem 23.2 (Graph Composition Satisfies Laws (kernel/CategoryBridge.v) — **(S)**)**.** *The operation `graph_compose_morphisms` satisfies associativity and unitality. Additionally:*

`categorical_extension_nofi_consistent`

MORPH and COMPOSE can carry $\Delta\mu = 0$ and are not in the mandatory-cost class. The NoFI policy enters through `MORPH_ASSERT`: asserting that a built morphism chain exists and activating the cert channel always costs $S(\Delta\mu) \geq 1$. Building the chain is free; certifying it never is.

Theorem 23.3 (Tensor Bifunctoriality (kernel/CategoryMonoidal.v) — **(S)**)**.** *`graph_tensor_morphisms` is a bifunctor: it satisfies the interchange law*

$$(f \otimes g); (h \otimes k) = (f; h) \otimes (g; k)$$

for composable, disjoint pairs.

23.1.4 The Categorical Separation Theorem

Theorem 23.4 (Categorical Separation (kernel/PartitionSeparation.v, §10) — **(S)**)**.** *There exist states s_1, s_2 such that:*

$$s_1.\mathtt{vm\_regs} = s_2.\mathtt{vm\_regs} \land s_1.\mathtt{vm\_mem} = s_2.\mathtt{vm\_mem} \land s_1.\mathtt{vm\_mu} = s_2.\mathtt{vm\_mu} \land s_1.\mathtt{vm\_err} = s_2.\mathtt{vm\_err}$$

but $s_1.\mathtt{vm\_graph.pg\_morphisms} \neq s_2.\mathtt{vm\_graph.pg\_morphisms}$.

Proof. Construct s_1 with `pg_morphisms` $= [(0, ms)]$ for any morphism state ms, and s_2 with `pg_morphisms` $= []$. Arrange identical `vm_regs`, `vm_mem`, `vm_mu`, `vm_err` by choosing costs and instruction sequences accordingly. Distinction by `discriminate`. □ □

Remark 23.1 (Significance)**.** A classical machine (exposing only registers, memory, and a cost counter) cannot distinguish s_1 from s_2. The Thiele Machine distinguishes them in one probe instruction: `MORPH_DELETE` 0 succeeds on s_1 and errors on s_2. The categorical layer records *how* knowledge was derived, not just *what* the final values are.

This is the machine's claim to being a *knowledge auditor*: any party claiming to have verified a structural relation must carry a μ-receipt $\geq$ the minimum construction cost. The receipt is unforgeable by the NoFI theorem applied to `MORPH_ASSERT`.

Chapter 24

Falsifiable Predictions

The honest falsifiable predictions are the structural theorems that compile or fail. The claims below identify specific mathematical conditions that would constitute counterexamples to the machine's core theorems.

24.1 Falsify No Free Insight

Find a Thiele Machine program trace that starts with `vm_certified = false` and $\mu = 0$, ends with `vm_certified = true`, and ends with $\mu = 0$. The Coq proof of `certification_requires_positive_mu` would fail.

24.2 Falsify the Classical CHSH Bound

Exhibit a locally factorizable strategy achieving $|S| > 2$. This contradicts `chsh_stat_violation_not_local` and also contradicts known mathematics.

24.3 Falsify the Biconditional

Exhibit $E_{00}, E_{01}, E_{10}, E_{11}$ satisfying the three column-contractivity conditions but with a non-PSD NPA moment matrix. The theorem `column_contractive_iff_quantum_realizable` says no such example exists.

24.4 Falsify the μ-Hierarchy

Find $k \geq 1$ and a trace reaching `vm_certified = true` and `vm_mu` $\geq k$ with total trace cost $< k$. This contradicts `mu_hierarchy_theorem`.

24.5 What I Don't Know

- **Physical interpretation of μ**: Whether the μ ledger corresponds to thermodynamic entropy in a physical Thiele Machine is an open empirical question. The formal development is consistent with Landauer's principle. The calibration hypothesis is named (`mu_landauer_unruh_calibrated`) and not proved in Coq.
- **Quantum mechanics connection**: The mathematical equivalence between Thiele honesty and zero-marginal NPA realizability is a kernel-tier theorem. Whether a physical Thiele Machine can produce CHSH-violating statistics by exploiting quantum entanglement in hardware is an open empirical question outside the scope of this project.

- **Complexity implications**: Whether $\mu\mathrm{P} \neq \mu\mathrm{NP}$ is an open question. The factored-search witness shows a concrete structural advantage but is not a class separation.

- **Physical constants**: No physical constant is derived from first principles. There is no derivation of Planck's constant h; any relation tying an internal time-scale τ_μ to h would be circular by construction, defining τ_μ in terms of h and checking consistency rather than predicting h.

- **The two +1s** (OP-Plus-One): the `+1` in `triangle_angle`'s denominator (`PI · d_bc / (1 + d_ab + d_ac + d_bc)`, `coq/kernel/curvature/MuGravity.v`) versus the A2 cost-floor `+1` on CERTIFY/LASSERT (`coq/kernel/foundation/VMStep.v`) — same axiom, or unrelated regularisation? No kernel-internal theorem unifies them without circularity (`coq/kernel/frontier/F3_PlusOneStructural.v`); the symbolic correction term decays as $-\pi/(3(3d+1))$ in the equilateral case (`tools/triangle_angle_plus_one_analysis.py`), regularisation-shaped rather than A2-quantum-shaped. Inconclusive. A positive structural identification (Case A, the geometric `+1` is A2 in geometric form) would re-frame the calibrated regime's asymptotic-only character as a structural consequence of A2; a positive non-identification (Case B) would license replacing $S(\dots)$ with $(\dots)$ in the denominator. Either is real categorical content. The kernel was not modified pending closure.

- **F3 narrow non-vacuity** (related to OP-Plus-One): a constructible finite `VMState` satisfying $\forall m$, `calibration_residual`$(s, m) = 0$ on a partition graph with at least one actual graph triangle and `well_formed_triangulated` holding. The symbolic search to $N = 7$ exhausts negatively. Whether the asymptotic-only character of calibration is a structural consequence of OP-Plus-One's Case A, or just a search-depth artifact, is itself open.

Appendix A

Proof File Index

The table below maps each theorem to its Coq source file. Every path listed is a file that exists in the active kernel tree (`coq/`) and compiles to a `.vo` on the current build. Concepts that previously appeared here under speculative paths (e.g. `theory/`, `thielemachine/coqproofs/`, `quantum_derivation/`, `physics_exploration/`) have been removed: they were aspirational citations from an earlier project layout that never made it into the active corpus, and the audit script (`scripts/audit_thesis_citations.py`) now flags such citations on every commit.

Result	Coq Source
Cost ledger and No Free Insight	
No Free Insight (Thm. 5.1)	`coq/nofi/NoFreeInsight_Theorem.v`
μ-Chaitin	`coq/nofi/MuChaitinTheory_Theorem.v`
μ-Conservation	`coq/kernel/foundation/MuLedgerConserva`
Receipt Integrity	`coq/kernel/nfi/ReceiptIntegrity.v`
Noether / Gauge	`coq/kernel/curvature/KernelNoether.v`
Strict Subsumption	`coq/kernel/foundation/Subsumption.v`
Universal NFI (any substrate)	`coq/kernel/nfi/UniversalCertificationC`
Quantitative NFI ($\geq S(d)$ trace bound)	`coq/kernel/mu_calculus/QuantitativeNoF`
μ-Hierarchy	`coq/kernel/mu_calculus/MuHierarchyTheo`
Partition Ops Free, Certification Non-Free	`coq/kernel/nfi/PartitionRefinementNoFI`
μ-Ledger Necessity (5-way classification)	`coq/NecessityOfMuLedger.v`, `coq/kernel/n`
Latent μ Intrinsic to Computation	`coq/NecessityOfMuLedger.v`: `shadow_mu_is`
μ Cost Universal Across Machines	`coq/NecessityOfMuLedger.v`: `shadow_mu_de`
μ Cost Unique Across Accounting Systems	`coq/NecessityOfMuLedger.v`: `shadow_mu_un`
No Free Computation (inevitable μ)	`coq/NecessityOfMuLedger.v`: `shadow_mu_in`
μ-budget predicates + zero-μ inclusion	`coq/kernel/mu_calculus/MuComplexity.v`
Categorical and shadow-projection layer	
Categorical Separation / Category Laws	`coq/kernel/category/CategoryLaws.v`, `coq`
Shadow Strictly Lossy	`coq/kernel/witness/ShadowProjection.v`:
Turing Strictness ($D5$)	`coq/kernel/foundation/TuringStrictness`
CHSH Coupling Bridge ($S \leq 2$ from local coupling, $S > 2$ rules it out)	`coq/kernel/quantum/CHSHCouplingBridge.`
Witness Insight Three-Tier Taxonomy	`coq/kernel/witness/WitnessInsightGener`
Quantum / CHSH algebra (zero-axiom)	
Thiele honest $\iff$ NPA realizable (biconditional)	`coq/kernel/quantum/QuantumPartitionPSD`
Forward: column-contractive $\Rightarrow$ NPA PSD	`coq/kernel/nfi/MuLedgerQuantumBridge.v`
Reverse: NPA PSD $\Rightarrow$ column-contractive	`coq/kernel/quantum/QuantumPartitionPSD`
Quadratic-form discriminant lemma	`coq/kernel/quantum/ConstructivePSD.v`: c
Tsirelson from column contractivity ($\lvert S\rvert \leq 2\sqrt{2}$)	`coq/kernel/nfi/MuLedgerQuantumBridge.v`
Tsirelson (algebraic, tight)	`coq/kernel/category/AlgebraicCoherence`
Tsirelson (algebraic from coherence)	`coq/kernel/quantum/TsirelsonFromAlgebr`
General Algebraic Tsirelson ($S^2 \leq 8$)	`coq/kernel/category/AlgebraicCoherence`
Conditional physics derivations (under named axioms; (C) tier)	
Born Rule	`coq/kernel/quantum/BornRule.v`
Unitarity	`coq/kernel/quantum/Unitarity.v`
Purification	`coq/kernel/quantum/Purification.v`
No-Cloning	`coq/kernel/quantum/NoCloning.v`
Information Causality	`coq/kernel/quantum/InformationCausalit`
Landauer derivation	`coq/thermodynamic/LandauerDerived.v`
Affine Off-Diagonal Ricci Zero (vertex 1)	`coq/kernel/curvature/AffineEFEClosure.`
Full Tensor EFE at Vertex 1 ($G = 3T$)	`coq/kernel/curvature/AffineEFEClosure.`
Outer Vertex Characterization (Ricci $= 3/32$)	`coq/kernel/curvature/AffineEFEClosure.`
Spacetime Emergence (consistency check)	`coq/kernel/curvature/SpacetimeEmergenc`
μ-Geometry (pseudometric)	`coq/kernel/mu_calculus/MuGeometry.v`
4D Simplicial Complex	`coq/kernel/curvature/FourDSimplicialCo`
Metric from μ-Costs	`coq/kernel/curvature/MetricFromMuCosts`
Riemann / Christoffel / Einstein tensors	`coq/kernel/curvature/RiemannTensor4D.v`
Stress-Energy Tensor / Conservation / Vacuum EFE	`coq/kernel/curvature/EinsteinEquations`
Einstein Equations (general, full diagonal-and-offdiag chain)	`coq/kernel/curvature/EinsteinEquations`
`A_QM` bridge axiom + master Tsirelson conditional	`coq/PhysicsConditionalClosure.v`
Self-reference, falsifiability, isomorphism	
Self-Reference Tower	`coq/self_reference/SelfReference.v`
Thiele Manifold	`coq/thiele_manifold/ThieleManifold.v`
Physical Constants (definitions only)	`coq/thiele_manifold/PhysicalConstants.`
Falsifiable Prediction record	`coq/kernel/aggregators/FalsifiablePred`
Three-Layer (Coq $\leftrightarrow$ Python $\leftrightarrow$ RTL) Isomorphism	`coq/kernel/hardware_bridge/ThreeLayerI`
Hardware / RTL / extraction	
RTL Gap Count $= 0$	`coq/kami_hw/RTLGapRegistry.v`: `rtl_gap_c`
RTL Coverage Partition ($37 + 10 + 0 = 47$)	`coq/kami_hw/RTLGapRegistry.v`: `rtl_cover`
RTL Trust Boundary (Section Variable)	`coq/kernel/hardware_bridge/VerilogRTLC`
BSC Compilation Trust Boundary	`coq/kernel/hardware_bridge/VerilogRTLC`
Coq-Kami refinement (proves `rtl_step_correct` for COQKAMI)	`coq/kami_hw/VerilogSemantics.v`

Appendix B

Epistemological Classification

The following table classifies every major result by its epistemological status.

Result	**Status**	**Key Assumption**
Unconditional structural theorems about the machine		
No Free Insight	**(S)**	Module type contract
μ-Chaitin	**(S)**	Module type contract
Strict Subsumption	**(S)**	Partition structure
Oracle Cost (instantiation of universal NFI)	**(S)**	Definitional (soundness := cost ≥ 1)
μ-Conservation	**(S)**	Machine semantics only
μ-Ledger Necessity	**(S)**	Machine semantics + explicit witnesses
μ-Ledger Minimality	**(S)**	Named projection lattice in `NecessityAbstract.`
Receipt Integrity	**(S)**	Hash-chain model
Gauge / Noether	**(S)**	Definitional (record structure)
Three-Layer Isomorphism / comparison contract	**(C)**	Backend decoders satisfy the shared observable pr
Discrete Angle-Defect Identity (planar triangulations)	**(S)**	Well-formed triangulated partition graph; equilate
Curvature from μ-Gradients (uniform mass $\Rightarrow G_{\mu\nu} = 0$)	**(R)**	Reference only; $0 = 0$ in vacuum
Vacuum Implies Zero Stress-Energy	**(R)**	Reference only; $0 = 0$ in vacuum
Einstein Equations (Vacuum)	**(R)**	Reference only; not a claim about physics; $G = 1$
Einstein Equations (General full tensor)	**(C)**	Named off-diagonal Ricci premise in `EinsteinEqu`
Affine Off-Diagonal Ricci Zero (vertex 1)	**(R)**	Reference only; specific boundary 4-simplex confi
Full Tensor EFE at Vertex 1 ($G = 3T$)	**(R)**	Reference only; not a physical claim
Outer Vertex Characterization	**(R)**	Reference only
Discrete Differential Geometry	**(R)**	Definitions only; reference exploration
Partition Ops Free / Certification Non-Free	**(S)**	Machine semantics only
CHSH Trial Is Tier 0	**(S)**	Machine semantics only
Three-Tier Observation Taxonomy	**(S)**	Machine semantics only
Locally Consistent $\Rightarrow$ Separable Coupling	**(S)**	Coupling definition
Local Coupling Bound ($\lvert S\rvert \leq 2$)	**(S)**	Coupling definition
$S > 2$ Rules Out Local Couplings	**(S)**	Coupling definition
RTL Gap Count $= 0$	**(S)**	Machine semantics + hardware spec
Thiele Honest $\iff$ NPA Realizable	**(S)**	Pure polynomial arithmetic; zero axioms
PR Box Not Thiele-Honest	**(S)**	Direct arithmetic on condition 1
Tsirelson Is Thiele Boundary (forward + contrapositive)	**(S)**	From column contractivity; converse not claimed
Classical $P \subseteq \mu P$	**(S)**	μ-complexity definitions
Conditional derivations (valid given stated axioms)		
Born Rule Uniqueness	**(C)**	Linearity assumption (`coq/kernel/quantum/Born`
Unitarity from $\mu = 0$	**(C)**	Info conservation (one direction only)
No-Cloning	**(C)**	Purity = information
Purification	**(C)**	Bloch sphere model
Landauer	**(C)**	Second law *assumed* as record field
Self-Reference	**(C)**	Dimension = complexity
Tsirelson Bound (algebraic)	**(C)**	Algebraic derivation (`TsirelsonFromAlgebra.v`,
Consistency relations (definitions verified, not predictions)		
Information Causality	**(R)**	Circular: equivalence assumed as hypothesis
μ-Geometry	**(R)**	Pseudometric (absolute value on $\mathbb{Z}$)
Spacetime Locality	**(R)**	Data isolation in partition graph
Born Rule (`BornRule.v`)	**(R)**	μ-consistent hypothesis unused

Legend: **(S)** = unconditional theorem about the machine; **(C)** = valid derivation conditional on stated axiom; **(R)** = internal consistency check.